

Novel fractional wavelet transform: Principles, MRA and application

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ABSTRACT

Wavelet transform (WT) can be viewed as a differently scaled bandpass filter in the frequency domain, so WT is not the optimal time-frequency representation method for those signals which are not band-limited in the frequency domain. A novel fractional wavelet transform (FRWT) is proposed to break the limitation of WT, it displays the time and fractional frequency information jointly in the time-fractional-frequency (TFF) plane. The definition and basic properties of FRWT are studied firstly. Furthermore, the multiresolution analysis and orthogonal fractional wavelets associated with FRWT are explored. Finally, the application of FRWT in the LFM signal TFF analysis is discussed and verified by simulations. The experimental results show that the energy concentration of LFM signal representation by proposed FRWT is better than that of some existing methods. The better energy concentration makes it can be further applied to the denoising, detection, parameter estimation and separation of LFM signal.

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1. Introduction

Nonstationary signals are widely used in radar, communication, navigation and other fields, time-frequency (TF) analysis is a powerful tool for analyzing and processing non-stationary signal. TF analysis method can provide the joint distribution information of time and frequency domain, and clearly describes the law of signal frequency changing with time [1]. As a conventional TF analysis method, wavelet transform (WT) can provide a “time-frequency” window that changes with frequency, which can adaptively meet the requirements of TF analysis [2,3]. Since each wavelet component can be viewed as a band-pass filter with different scales in the frequency domain, the signal analysis ability of WT is limited in the time-frequency plane. Therefore, WT is not the optimal TF analysis method for the signal which is not well concentrated in the frequency domain. For example, linear frequency modulation (LFM) signal [4,5] is this kind of typical non-stationary signal, its amplitude spectrum in the frequency domain is approximately rectangular with a large frequency bandwidth [6].

Fractional Fourier transform (FRFT) is a generalized form of Fourier transform, it can represent signals in any fractional domain between time domain and frequency domain [7,8]. Fractional WT (FRWT) [9–15] is proposed inspired by the characteristic of FRFT, it can be viewed as a band-pass filter with different scales in

the fractional domain. FRWT successfully breaks the limitation of WT, and has been applied to LFM signal separation and denoising [11], image fusion [16], vibrator data denoising [17]. The earliest FRWT was proposed by Mendlovic et al. [9], it first performs a FRFT with the optimal fractional order over the entire input signal and then performs WT on each fractional spectrum. Therefore, this FRWT can not obtain information about the local characteristics of input signal. Recently, a new FRWT was proposed by Shi et al. with fractional convolution, it can offer signal representations in the time-fractional-frequency plane [10–12]. However, the fractional domain window in [10] is still a function associated with FT. Dai defined a new type of FRWT [13], this FRWT has more elegant mathematical properties and is more general than FRWT in [11]. Although this new FRWT uses fractional window to overcome the drawback of FRWT in [11], the window and signal are restricted to analysis in the same fractional domain. This restriction may cause energy dispersion in the signal representation.

On the basis of above analysis, this paper aims to define a novel FRWT to break the limitations of WT and existing FRWT. Compared with WT and existed FRWT, this novel FRWT has following characteristics: (1) FRWT can be viewed as a differently scaled bandpass filter in the fractional domain, it not only breaks the limitation of WT, but also overcomes the defect of FRFT in the signal local characteristics representation; (2) this FRWT uses the fractional window, which has wider applicability over the FRWT in [10]; (3) the window and signal can be transformed freely in different fractional domains, which provides greater flexibility in comparison with the FRWT in [13]. The main contributions of this paper in-

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clude: (1) a novel FRWT is proposed to overcome the limitations of WT and some existing FRWT, its definition and basic properties are established; (2) the MRA and construction of orthogonal fractional wavelet for FRWT are deduced; (3) the application of FRWT in the LFM signal time-fractional-frequency (TFF) analysis is explored.

The remaining sections of this paper are organized as follows: the definition of FRFT and fractional convolution theorem are firstly presented in Section 2. Moreover, the definition and basic properties of FRWT are established in Section 3. The MRA and construction of orthogonal fractional wavelet for FRWT are deduced in section 4. In section 5, the application of FRWT in the LFM signal TFF analysis is explored. Finally, the conclusion is stated in Section 6.

2. Preliminaries

2.1. Fractional Fourier transform

Definition 1. The fractional Fourier transform (FRFT) [7] with parameter α of $f(t)$ is defined as

$$F_f^\alpha(u) = \mathcal{F}^\alpha[f(t)](u) = \int_{\mathbb{R}} f(t) K_\alpha(u, t) dt \quad (1)$$

with the kernel

$$K_\alpha(u, t) = \begin{cases} A_\alpha e^{\frac{i}{2}[(t^2+u^2)\cot\alpha - 2tucsc\alpha]}, & \alpha \neq k\pi \\ \delta(t-u), & \alpha = 2k\pi \\ \delta(t+u), & \alpha = (2k-1)\pi \end{cases} \quad (2)$$

where $A_\alpha = \sqrt{(1-i\cot\alpha)/2\pi}$, $F_f^\alpha(u)$ and $\mathcal{F}^\alpha$ denote the FRFT of $f(t)$ and FRFT operator respectively. The relationship between FRFT and FT is [11]

$$\mathcal{F}^\alpha[f(t)](u) = \sqrt{1-i\cot\alpha} e^{\frac{i}{2}u^2\cot\alpha} \mathcal{F}[f(t)e^{\frac{i}{2}t^2\cot\alpha}](ucsc\alpha) \quad (3)$$

This shows that FRFT reduces to FT when $\alpha = \pi/2$.

Lemma 1. The relationship between WVD_f and $WVD_{F_f^\alpha}$ is [18]

$$WVD_{F_f^\alpha}(u, v) = WVD_f(u \cos \alpha - v \sin \alpha, u \sin \alpha + v \cos \alpha) \quad (4)$$

where $WVD_f(u, v)$ denotes the Wigner-Ville distribution (WVD) of $f(t)$, its definition is

$$WVD_f(u, v) = \int_{\mathbb{R}} f(u + \tau/2) \overline{f(u - \tau/2)} e^{-iv\tau} d\tau \quad (5)$$

It shows that FRFT can be regarded as the rotation of time-frequency plane, and thus FRFT can provide a fractional domain signal representation between time domain and frequency domain.

Lemma 2. The discrete time FRFT (DTFRFT) of a sequence $f_n \in \ell^2(\mathbb{Z})$ has the chirp periodicity [11], i.e.,

$$\tilde{F}_{f_n}^\alpha(u + 2k\pi \sin \alpha) e^{-\frac{i}{2}(u+2k\pi \sin \alpha)^2 \cot \alpha} = \tilde{F}_{f_n}^\alpha(u) e^{-\frac{i}{2}u^2 \cot \alpha} \quad (6)$$

where $\tilde{F}_{f_n}^\alpha(u)$ denotes the DTFRFT of f_n , its definition is

$$\tilde{F}_{f_n}^\alpha(u) = \tilde{F}^\alpha[f_n](u) = \sum_{n \in \mathbb{Z}} f_n K_\alpha(n, u) \quad (7)$$

Lemma 3. The inner product theorem associated with FRFT is [7]

$$\int_{\mathbb{R}} f(t) \overline{g(t)} dt = \int_{\mathbb{R}} F_f^\alpha(u) \overline{F_g^\alpha(u)} du \quad (8)$$

In particularly, if $f(t) = g(t)$, then

$$\int_{\mathbb{R}} |f(t)|^2 dt = \int_{\mathbb{R}} |F_f^\alpha(u)|^2 du \quad (9)$$

2.2. Fractional convolution

Definition 2. The generalized fractional convolution of $f(t), g(t) \in L^2(\mathbb{R})$ is given by [19]

$$\begin{aligned} & g(t) \Theta_{\alpha, \beta, \gamma} h(t) \\ &= \int_{-\infty}^{+\infty} h(\tau) g(t - \tau) e^{-\frac{i}{2}[\tau^2(\cot\alpha - \cot\gamma) - 2t\tau \cot\alpha + t^2(\cot\beta - \cot\alpha)]} d\tau \end{aligned} \quad (10)$$

where $\Theta_{\alpha, \beta, \gamma}$ denotes the generalized fractional convolution operator. The classical convolution and some existing earlier fractional convolution definitions related to FRFT are shown to be special cases of Definition 2, the details can be found in the Ref. [19]. The relationship between generalized fractional convolution and classical convolution is given by

$$g(t) \Theta_{\alpha, \beta, \gamma} h(t) = e^{-\frac{i}{2}t^2 \cot \alpha} [g(t) e^{\frac{i}{2}t^2 \cot \beta} * h(t) e^{\frac{i}{2}t^2 \cot \gamma}] \quad (11)$$

where $*$ denotes the classical convolution operator.

Lemma 4. According to Eqs. (1) and (10), the fractional convolution theorem can be expressed as [19]

$$\mathcal{F}^\alpha[g(t) \Theta_{\alpha, \beta, \gamma} h(t)](u) = \epsilon_{\alpha, \beta, \gamma}(u) F_g^\beta\left(\frac{ucsc\alpha}{csc\beta}u\right) F_h^\gamma\left(\frac{ucsc\alpha}{csc\gamma}u\right) \quad (12)$$

where

$$\epsilon_{\alpha, \beta, \gamma}(u) = \frac{A_\alpha}{A_\beta A_\gamma} e^{\frac{i}{2}u^2[\cot\alpha - (\frac{csc\alpha}{csc\beta})^2 \cot\beta - (\frac{csc\alpha}{csc\gamma})^2 \cot\gamma]} \quad (13)$$

3. Novel fractional wavelet transform

3.1. Definition

Definition 3. A novel fractional wavelet transform (FRWT) of $f(t)$ is defined as

$$\begin{aligned} W_f^{\alpha, \beta}(a, b) &= \mathcal{W}^{\alpha, \beta}[f(t)](a, b) = \int_{\mathbb{R}} f(t) \overline{\psi_{a, b, \alpha, \beta}(t)} dt \\ &= \frac{1}{\sqrt{a}} \int_{\mathbb{R}} f(t) \overline{\psi\left(\frac{t-b}{a}\right)} e^{\frac{i}{2}[(t^2-b^2)\cot\alpha - (\frac{t-b}{a})^2 \cot\beta]} dt \end{aligned} \quad (14)$$

with

$$\psi_{a, b, \alpha, \beta}(t) = \frac{1}{\sqrt{a}} \psi\left(\frac{t-b}{a}\right) e^{-\frac{i}{2}[(t^2-b^2)\cot\alpha - (\frac{t-b}{a})^2 \cot\beta]} \quad (15)$$

where $\psi(\frac{t-b}{a}) (a \in \mathbb{R}^+, b \in \mathbb{R})$ is the continuous wavelet function. This FRWT is a unified form of some existing FRWT, some special cases are listed as follows:

- 1) when $\alpha = \beta = \pi/2$, this FRWT reduces to classical WT;
- 2) when $\alpha = \beta$, this FRWT reduces to FRWT proposed by Dai [13], i.e.,

$$W_f^\alpha(a, b) = \frac{1}{\sqrt{a}} \int_{\mathbb{R}} f(t) \overline{\psi\left(\frac{t-b}{a}\right)} e^{\frac{i}{2}[(t^2-b^2)\cot\alpha - (\frac{t-b}{a})^2\cot\alpha]} dt \quad (16)$$

- 3) when $\beta = \pi/2$, this FRWT reduces to FRWT proposed by Shi [10], i.e.,

$$W_f^\alpha(a, b) = \frac{1}{\sqrt{a}} \int_{\mathbb{R}} f(t) \overline{\psi\left(\frac{t-b}{a}\right)} e^{\frac{i}{2}(t^2-b^2)\cot\alpha} dt \quad (17)$$

Based on the fractional convolution, this FRWT can be rewritten as

$$W_f^{\alpha,\beta}(a, b) = f(t) \Theta_{\alpha,\alpha,\beta} \frac{1}{\sqrt{a}} \overline{\psi\left(-\frac{t}{a}\right)} \quad (18)$$

Therefore, FRWT can be viewed as the fractional convolution of signal $f(t)$ and $\frac{1}{\sqrt{a}} \overline{\psi\left(-\frac{t}{a}\right)}$ in the time domain. According to Eq. (11), the continuous FRWT can be computed by

$$W_f^{\alpha,\beta}(a, b) = e^{-\frac{i}{2}b^2\cot\alpha} [f(t) e^{\frac{i}{2}t^2\cot\alpha} * \frac{1}{\sqrt{a}} \overline{\psi\left(-\frac{t}{a}\right)} e^{\frac{i}{2}t^2\cot\beta}] \quad (19)$$

Compared with continuous WT, the computation of continuous FRWT only adds three products with chirp signal. It indicates that the computational complexity of FRWT depends on the computational complexity of WT, the computational complexity of WT is $O(N)$, where N is the length of signal. Therefore, the computational complexity of proposed FRWT is $O(N)$, which is identical with that of FRWT in [10] and [13].

Since

$$\mathcal{F}^\alpha[\psi_{a,b,\alpha,\beta}(t)](u) = \frac{\sqrt{a}}{A_\beta} e^{-\frac{i}{2}(au/\lambda)^2\cot\beta} F_\psi^\beta(au/\lambda) K_\alpha(u, b) \quad (20)$$

Then, from Eqs. (8) and (14), FRWT can also be expressed in term of the FRFT, i.e.,

$$W_f^{\alpha,\beta}(a, b) = \frac{\sqrt{a}}{A_\beta} \int_{\mathbb{R}} e^{\frac{i}{2}(au/\lambda)^2\cot\beta} F_f^\alpha(u) \overline{F_\psi^\beta(au/\lambda)} K_{-\alpha}(u, b) du \quad (21)$$

where $\lambda = \csc\beta/\csc\alpha$. Eq. (21) states that each fractional wavelet component can be viewed as a scaled bandpass filter in the fractional domain. This means that FRWT breaks the limitation of traditional wavelet analysis in time-frequency plane, and overcomes the defect of FRFT in the signal local characteristics representation. According to Eqs. (18) and (21), the differences between proposed FRWT and FRWT in [13] mainly include: 1) in terms of time domain, the proposed FRWT is defined by a different fractional convolution. The window function $\psi_{\alpha,\beta,a,b}$ of proposed FRWT has one more free parameter β , and thus the proposed FRWT has more freedom of parameter selection; 2) in terms of fractional domain, the window and signal are restricted to same fractional domain (i.e., F_ψ^α and F_f^α) for FRWT in [13]. However, the proposed FRWT breaks the restriction of FRWT in [13], the window and signal can be transformed freely to different fractional domains (i.e., F_ψ^β and F_f^α).

3.2. Basic properties

1) Linear property

If $f_1(t) \rightarrow W_{f_1}^{\alpha,\beta}(a, b)$, $f_2(t) \rightarrow W_{f_2}^{\alpha,\beta}(a, b)$, and $f(t) = k_1 f_1(t) + k_2 f_2(t)$, then

$$W_f^{\alpha,\beta}(a, b) = k_1 W_{f_1}^{\alpha,\beta}(a, b) + k_2 W_{f_2}^{\alpha,\beta}(a, b) \quad (22)$$

2) Scaling property

If $f(t) \rightarrow W_f^{\alpha,\beta}(a, b)$, $g(t) = f(ct)$, then

$$W_g^{\alpha,\beta}(a, b) = \frac{1}{\sqrt{c}} W_f^{\alpha',\beta}(ca, cb) \quad (23)$$

where $c > 0$ and α' satisfying $\cot\alpha' = \cot\alpha/c^2$.

Proof. From Eq. (14), we have

$$\begin{aligned} W_g^{\alpha,\beta}(a, b) &= \frac{1}{\sqrt{a}} \int_{\mathbb{R}} f(ct) \overline{\psi\left(\frac{t-b}{a}\right)} e^{\frac{i}{2}[(t^2-b^2)\cot\alpha - (\frac{t-b}{a})^2\cot\beta]} dt \\ &= \frac{1}{\sqrt{c}} \int_{\mathbb{R}} f(\xi) \overline{\psi\left(\frac{\xi-cb}{ca}\right)} e^{\frac{i}{2}[(\xi^2-c^2b^2)\frac{\cot\alpha}{c^2} - (\frac{\xi-cb}{ca})^2\cot\beta]} d\xi (\xi = ct) \\ &= \frac{1}{\sqrt{c}} W_f^{\alpha',\beta}(ca, cb) \end{aligned}$$

3) Inner product theorem

If $f(t) \rightarrow W_f^{\alpha,\beta}(a, b)$, $g(t) \rightarrow W_g^{\alpha,\beta}(a, b)$, and $\psi(t)$ satisfies the “admissibility” condition, i.e.,

$$C_{\psi,\beta} = \int_{-\infty}^{+\infty} \frac{|F_\psi^\beta(\Omega)|^2}{\Omega} d\Omega < \infty, \quad (24)$$

then,

$$\int_{\mathbb{R}^2} W_f^{\alpha,\beta}(a, b) \overline{W_g^{\alpha,\beta}(a, b)} \frac{1}{a^2} da db = 2\pi \sin\beta C_{\psi,\beta} \int_{\mathbb{R}} f(t) \overline{g(t)} dt. \quad (25)$$

Proof. From Eq. (21), we have

$$W_f^{\alpha,\beta}(a, b) = \frac{\sqrt{a}}{A_\beta} \int_{\mathbb{R}} e^{-\frac{i}{2}(au/m)^2\cot\beta} F_f^\alpha(u) \overline{F_\psi^\beta(au/m)} K_{-\alpha}(u, b) du \quad (26)$$

$$W_g^{\alpha,\beta}(a, b) = \frac{\sqrt{a}}{A_\beta} \int_{\mathbb{R}} e^{-\frac{i}{2}(av/m)^2\cot\beta} F_g^\alpha(v) \overline{F_\psi^\beta(av/m)} K_{-\alpha}(v, b) dv \quad (27)$$

Owing to

$$\int_{\mathbb{R}} K_{-\alpha}(u, b) \overline{K_{-\alpha}(v, b)} db = \delta(u - v) \quad (28)$$

Then, we have

$$\begin{aligned}
& \int_{\mathbb{R}^2} W_f^{\alpha,\beta}(a,b) \overline{W_g^{\alpha,\beta}(a,b)} \frac{1}{a^2} da db \\
&= 2\pi \sin \beta \int_{\mathbb{R}} \frac{|F_\psi^\beta(au/m)|^2}{a} da \int_{\mathbb{R}} F_f^\alpha(u) \overline{F_g^\alpha(u)} du \\
&= 2\pi \sin \beta C_{\psi,\beta} \int_{\mathbb{R}} F_f^\alpha(u) \overline{F_g^\alpha(u)} du \\
&= 2\pi \sin \beta C_{\psi,\beta} \int_{-\infty}^{+\infty} f(t) \overline{g(t)} dt
\end{aligned}$$

In particular, let $f(t) = g(t)$, then the Parseval's identity associated with FRWT is

$$\int_{\mathbb{R}^2} |W_f^{\alpha,\beta}(a,b)|^2 \frac{1}{a^2} da db = 2\pi \sin \beta C_{\psi,\beta} \int_{-\infty}^{+\infty} |f(t)|^2 dt. \quad (29)$$

4) Inverse formula

Let $g(t) = \delta(t - \tau)$, from Eq. (25), the inverse formula of FRWT is given by

$$f(t) = \frac{1}{2\pi \sin \beta C_{\psi,\beta}} \int_{\mathbb{R}^2} W_f^{\alpha,\beta}(a,b) \psi_{a,b,\alpha,\beta}(t) \frac{1}{a^2} da db \quad (30)$$

where $C_{\psi,\beta} < \infty$ is the “admissibility” condition for $\psi(t)$.

5) Reproducing kernel

Suppose that (a_0, b_0) is an arbitrary point on the (a, b) plane, $W_f^{\alpha,\beta}(a, b)$ is the FRWT of some function under parameter α and β , then $W_f^{\alpha,\beta}(a, b)$ has to satisfy the following reproducing kernel equation

$$W_f^{\alpha,\beta}(a_0, b_0) = \int_{\mathbb{R}^2} W_f^{\alpha,\beta}(a, b) K_{\psi,\alpha,\beta}(a_0, b_0; a, b) \frac{1}{a^2} da db \quad (31)$$

where $K_{\psi,\alpha,\beta}(a_0, b_0; a, b)$ is the reproducing kernel, i.e.,

$$K_{\psi,\alpha,\beta}(a_0, b_0; a, b) = \frac{1}{2\pi \sin \beta C_{\psi,\beta}} \int_{\mathbb{R}} \psi_{a,b,\alpha,\beta}(t) \overline{\psi_{a_0,b_0,\alpha,\beta}(t)} dt \quad (32)$$

Proof. Based on Eqs. (14) and (30), then

$$\begin{aligned}
& W_f^{\alpha,\beta}(a_0, b_0) \\
&= \int_{\mathbb{R}} f(t) \overline{\psi_{a_0,b_0,\alpha,\beta}(t)} dt \\
&= \int_{\mathbb{R}} \left[\frac{1}{2\pi \sin \beta C_{\psi,\beta}} \int_{\mathbb{R}^2} W_f^{\alpha,\beta}(a, b) \psi_{a,b,\alpha,\beta}(t) \frac{1}{a^2} da db \right] \\
&\quad \cdot \overline{\psi_{a_0,b_0,\alpha,\beta}(t)} dt \\
&= \int_{\mathbb{R}^2} W_f^{\alpha,\beta}(a, b) \left[\frac{1}{2\pi \sin \beta C_{\psi,\beta}} \int_{\mathbb{R}} \psi_{a,b,\alpha,\beta}(t) \overline{\psi_{a_0,b_0,\alpha,\beta}(t)} dt \right] \\
&\quad \cdot \frac{1}{a^2} da db \\
&= \int_{\mathbb{R}^2} W_f^{\alpha,\beta}(a, b) K_{\psi,\alpha,\beta}(a_0, b_0; a, b) \frac{1}{a^2} da db
\end{aligned}$$

4. Multiresolution analysis and orthogonal fractional wavelets

4.1. Multiresolution analysis

Definition 4. An orthogonal MRA of $L^2(\mathbb{R})$ associated with FRWT is defined as a sequence of closed subspace $V_m^{\alpha,\beta} \subseteq L^2(\mathbb{R}) (m \in \mathbb{Z})$ such that

- (1) $V_m^{\alpha,\beta} \subseteq V_{m+1}^{\alpha,\beta}, \forall m \in \mathbb{Z};$
- (2) $f(t) \in V_m^{\alpha,\beta} \Leftrightarrow f(2t)e^{\frac{i}{2}[(2t)^2 - t^2] \cot \alpha} \in V_{m+1}^{\alpha,\beta}, \forall m \in \mathbb{Z};$
- (3) $\bigcap_{m \in \mathbb{Z}} V_m^{\alpha,\beta} = \{0\}, \bigcup_{m \in \mathbb{Z}} V_m^{\alpha,\beta} = L^2(\mathbb{R});$
- (4) There exists a function $\phi(t) \in L^2(\mathbb{R})$ such that $\{\phi_{\alpha,\beta,0,n} = \phi(t-n)e^{-\frac{i}{2}[(t^2 - n^2) \cot \alpha - (t-n)^2 \cot \beta]}\}_{n \in \mathbb{Z}}$ is an orthonormal basis of the subspace $V_0^{\alpha,\beta}$, where $\phi(t)$ is called scaling function of the given MRA.

Theorem 1. Assume $\{V_m^{\alpha,\beta}\}_{m \in \mathbb{Z}}$ is a orthogonal MRA of $L^2(\mathbb{R})$ for FRWT, $\phi(t)$ is the corresponding scaling function. For any $m \in \mathbb{Z}$, these functions

$$\{\phi_{\alpha,\beta,m,n}(t) = 2^{\frac{m}{2}} \phi(2^m t - n) e^{-\frac{i}{2}[(t^2 - (\frac{n}{2^m})^2) \cot \alpha - (2^m t - n)^2 \cot \beta]}\}_{n \in \mathbb{Z}} \quad (33)$$

form an orthonormal basis of the subspace $V_m^{\alpha,\beta}$.

Proof. Owing to

$$\begin{aligned}
& \langle \phi_{\alpha,\beta,m,n}, \phi_{\alpha,\beta,m,k} \rangle \\
&= 2^m e^{\frac{i}{2} 2^{-2m} (n^2 - k^2) \cot \alpha} \int_{\mathbb{R}} \phi(2^m t - n) e^{\frac{i}{2} (2^m t - n)^2 \cot \beta} \\
&\quad \cdot \overline{\phi(2^m t - k) e^{-\frac{i}{2} (2^m t - k)^2 \cot \beta}} dt \\
&= e^{\frac{i}{2} 2^{-2m} (n^2 - k^2) \cot \alpha} \int_{\mathbb{R}} \phi(\eta - n) e^{\frac{i}{2} (\eta - n)^2 \cot \beta} \\
&\quad \cdot \overline{\phi(\eta - k) e^{-\frac{i}{2} (\eta - k)^2 \cot \beta}} d\eta \quad (\text{let } \eta = 2^m t) \\
&= \delta(n - k)
\end{aligned}$$

where $\langle f, g \rangle = \int_{\mathbb{R}} f(t) \overline{g(t)} dt$ denotes the inner product of f and g . Hence, $\{\phi_{\alpha,\beta,m,n}(t)\}_{n \in \mathbb{Z}}$ form an complete orthonormal systems. Next, for any $f(t) \in V_m^{\alpha,\beta}$, according to the monotonicity of MRA, we have

$$f(2^{-m}t) e^{\frac{i}{2}[(2^{-m}t)^2 - t^2] \cot \alpha} \in V_0^{\alpha,\beta}. \quad (34)$$

Hence, $f(2^{-m}t) e^{\frac{i}{2}[(2^{-m}t)^2 - t^2] \cot \alpha}$ can be expressed as a combination of the basis $\{\phi_{\alpha,\beta,0,n}\}_{n \in \mathbb{Z}}$, i.e.,

$$\begin{aligned}
& f(2^{-m}t) e^{\frac{i}{2}[(2^{-m}t)^2 - t^2] \cot \alpha} \\
&= \sum_{n \in \mathbb{Z}} c_n \phi(t - n) e^{-\frac{i}{2}[(t^2 - n^2) \cot \alpha - (t-n)^2 \cot \beta]} \\
&\Rightarrow f(2^{-m}t) = \sum_{n \in \mathbb{Z}} c_n \phi(t - n) e^{-\frac{i}{2}[(2^{-m}t)^2 - n^2] \cot \alpha - (t-n)^2 \cot \beta} \\
&\Rightarrow f(t) = \sum_{n \in \mathbb{Z}} c'_n 2^{\frac{m}{2}} \phi(2^m t - n) e^{-\frac{i}{2}[(t^2 - (2^{-m}n)^2] \cot \alpha - (2^m t - n)^2 \cot \beta]}
\end{aligned}$$

where $c'_n = c_n 2^{-\frac{m}{2}} e^{\frac{i}{2}n^2(1-2^{-2m}) \cot \alpha}$. It shows that any $f(t) \in V_m^{\alpha,\beta}$ can be expressed as a combination of the $\{\phi_{\alpha,\beta,m,n}\}_{n \in \mathbb{Z}}$. Therefore,

the subspace $V_m^{\alpha,\beta}$ is generated by $\phi(t)$ with dilation, translation and chirp modulation, i.e.,

$$V_m^{\alpha,\beta} = \overline{\text{span}}\{\phi_{\alpha,\beta,m,n}(t)\} \\ = 2^{\frac{m}{2}} \phi(2^m t - n) e^{-\frac{i}{2}[(t^2 - (\frac{n}{2^m})^2) \cot \alpha - (2^m t - n)^2 \cot \beta]} \}_{n \in \mathbb{Z}} \quad (35)$$

If the orthogonality of MRA is relaxed by assuming that $\{\phi_{\alpha,\beta,0,n}\}_{n \in \mathbb{Z}}$ is a Riesz basis of $V_0^{\alpha,\beta}$, the $\phi(t)$ generates a generalized MRA of $L^2(\mathbb{R})$ associated with FRWT.

Theorem 2. The set of functions $\{\phi_{\alpha,\beta,0,n} = \phi(t - n) \times e^{-\frac{i}{2}[(t^2 - n^2) \cot \alpha - (t - n)^2 \cot \beta]}\}_{n \in \mathbb{Z}}$ is a Riesz basis of $V_0^{\alpha,\beta}$ if and only if there exist constants $0 < A \leq B < +\infty$ such that

$$A \leq \sum_{k \in \mathbb{Z}} |F_\phi^\beta(\frac{u + 2k\pi \sin \alpha}{\lambda})|^2 \leq B, \forall u \in [0, 2\pi \sin \alpha]. \quad (36)$$

Proof. Necessity: for any $f(t) \in V_0^{\alpha,\beta}$, we have

$$f(t) = \sum_{n \in \mathbb{Z}} c_n \phi(t - n) e^{-\frac{i}{2}[(t^2 - n^2) \cot \alpha - (t - n)^2 \cot \beta]} \quad (37)$$

By taking FRFT with α on both sides of Eq. (37), then

$$F_f^\alpha(u) = \mathcal{F}^\alpha \left[\sum_{n \in \mathbb{Z}} c_n \phi(t - n) e^{-\frac{i}{2}[(t^2 - n^2) \cot \alpha - (t - n)^2 \cot \beta]} \right] (u) \\ = \int_{\mathbb{R}} A_\alpha \sum_{n \in \mathbb{Z}} c_n \phi(t - n) e^{-\frac{i}{2}[(t^2 - n^2) \cot \alpha - (t - n)^2 \cot \beta]} \\ \cdot e^{\frac{i}{2}[(t^2 + u^2) \cot \alpha - 2tucsc\alpha]} dt \\ = \int_{\mathbb{R}} A_\alpha \sum_{n \in \mathbb{Z}} c_n \phi(t - n) e^{\frac{i}{2}[(u^2 + n^2) \cot \alpha + (t - n)^2 \cot \beta - 2tucsc\alpha]} dt \\ \stackrel{\xi = t - n}{=} \int_{\mathbb{R}} A_\alpha \sum_{n \in \mathbb{Z}} c_n \phi(\xi) e^{\frac{i}{2}[(u^2 + n^2) \cot \alpha + \xi^2 \cot \beta - 2(\xi + n)ucsc\alpha]} d\xi \quad (38)$$

$$= e^{-\frac{i}{2}(\frac{u}{\lambda})^2 \cot \beta} \int_{\mathbb{R}} \phi(\xi) e^{\frac{i}{2}[\xi^2 \cot \beta + (\frac{u}{\lambda})^2 \cot \beta - 2\xi \frac{u}{\lambda} \csc \beta]} d\xi \\ \cdot \sum_{n \in \mathbb{Z}} A_\alpha c_n e^{\frac{i}{2}[(u^2 + n^2) \cot \alpha - 2nucsc\alpha]} \\ = \frac{1}{A_\beta} e^{-\frac{i}{2}(\frac{u}{\lambda})^2 \cot \beta} \tilde{F}_{c_n}^\alpha(u) F_\phi^\beta(u/\lambda)$$

Owing to the Parseval's identity of FRFT (see Eq. (9)), we have

$$\|f(t)\|_{L^2}^2 = \|F_f^\alpha(u)\|_{L^2}^2 \\ = \frac{1}{|A_\beta|^2} \int_{\mathbb{R}} |\tilde{F}_{c_n}^\alpha(u) F_\phi^\beta(u/\lambda)|^2 du \\ = \frac{1}{|A_\beta|^2} \sum_{k \in \mathbb{Z}} \int_0^{2\pi \sin \alpha} |\tilde{F}_{c_n}^\alpha(u + 2k\pi \sin \alpha) F_\phi^\beta(\frac{u + 2k\pi \sin \alpha}{\lambda})|^2 du \\ = \frac{1}{|A_\beta|^2} \int_0^{2\pi \sin \alpha} |\tilde{F}_{c_n}^\alpha(u)|^2 \cdot \sum_{k \in \mathbb{Z}} |F_\phi^\beta(\frac{u + 2k\pi \sin \alpha}{\lambda})|^2 du$$

if $A \leq \sum_{k \in \mathbb{Z}} |F_\phi^\beta(\frac{u + 2k\pi \sin \alpha}{\lambda})|^2 \leq B$, combined with

$$\|c_n\|_{\ell^2}^2 = \int_0^{2\pi \sin \alpha} |\tilde{F}_{c_n}^\alpha(u)|^2 du, \quad (39)$$

then,

$$\frac{A}{|A_\beta|^2} \|c_n\|_{\ell^2}^2 \leq \|f(t)\|_{L^2}^2 = \sum_{n \in \mathbb{Z}} c_n \phi_{\alpha,\beta,0,n} \|_{L^2}^2 \leq \frac{B}{|A_\beta|^2} \|c_n\|_{\ell^2}^2 \quad (40)$$

Therefore, the set of functions $\{\phi_{\alpha,\beta,0,n}\}_{n \in \mathbb{Z}}$ is a Riesz basis of $V_0^{\alpha,\beta}$.

Sufficiency: if the set of functions $\{\phi_{\alpha,\beta,0,n}(t)\}_{n \in \mathbb{Z}}$ is a Riesz basis of $V_0^{\alpha,\beta}$, then

$$A_1 \|c_n\|_{\ell^2}^2 \leq \|f(t)\|_{L^2}^2 = \sum_{n \in \mathbb{Z}} c_n \phi_{\alpha,\beta,0,n} \|_{L^2}^2 \leq B_1 \|c_n\|_{\ell^2}^2 \quad (41)$$

Owing to

$$\|c_n\|_{\ell^2}^2 = \int_0^{2\pi \sin \alpha} |\tilde{F}_{c_n}^\alpha(u)|^2 du, \\ \|f(t)\|_{L^2}^2 = \frac{1}{|A_\beta|^2} \int_0^{2\pi \sin \alpha} |\tilde{F}_{c_n}^\alpha(u)|^2 \cdot \sum_{k \in \mathbb{Z}} |F_\phi^\beta(\frac{u + 2k\pi \sin \alpha}{\lambda})|^2 du$$

Hence, we have

$$A_1 \int_0^{2\pi \sin \alpha} |\tilde{F}_{c_n}^\alpha(u)|^2 du \\ \leq \frac{1}{|A_\beta|^2} \int_0^{2\pi \sin \alpha} |\tilde{F}_{c_n}^\alpha(u)|^2 \cdot \sum_{k \in \mathbb{Z}} |F_\phi^\beta(\frac{u + 2k\pi \sin \alpha}{\lambda})|^2 du \\ \leq B_1 \int_0^{2\pi \sin \alpha} |\tilde{F}_{c_n}^\alpha(u)|^2 du \quad (42)$$

Owing to the arbitrariness of $\tilde{F}_{c_n}^\alpha(u)$, then

$$A \leq \sum_{k \in \mathbb{Z}} |F_\phi^\beta(\frac{u + 2k\pi \sin \alpha}{\lambda})|^2 \leq B \text{ a.e. } \forall u \in [0, 2\pi \sin \alpha], \quad (43)$$

where $A = A_1 |A_\beta|^2$, $B = B_1 |A_\beta|^2$. In particular, $\{\phi_{\alpha,\beta,0,n}\}_{n \in \mathbb{Z}}$ is an orthonormal basis of $V_0^{\alpha,\beta}$ if and only if $A = B = \frac{1}{2\pi \sin \beta}$.

Theorem 3. Assume $\{V_m^{\alpha,\beta}\}_{k \in \mathbb{Z}}$ is a generalized MRA of $L^2(\mathbb{R})$ associated with FRWT, its scale function is $\phi(t)$. Let

$$F_\phi^\beta(u/\lambda) = \frac{F_\phi^\beta(u/\lambda)}{\sqrt{2\pi \sin \beta} G_{\phi,\alpha,\beta}(u)}, \quad (44)$$

where

$$G_{\phi,\alpha,\beta}(u) = \sqrt{\sum_{k \in \mathbb{Z}} |F_\phi^\beta(\frac{u + 2k\pi \sin \alpha}{\lambda})|^2} \quad (45)$$

Then, the set of functions $\{\phi_{\alpha,\beta,0,n} = \phi(t - n) \times e^{-\frac{i}{2}[(t^2 - n^2) \cot \alpha - (t - n)^2 \cot \beta]}\}_{n \in \mathbb{Z}}$ form an orthonormal basis of $V_0^{\alpha,\beta}$.

Proof. Let $\phi(t) e^{-\frac{i}{2}(\cot \alpha - \cot \beta)t^2} \in V_0^{\alpha,\beta}$, then

$$\phi(t) = \sum_{n \in \mathbb{Z}} c_n \phi(t - n) e^{\frac{i}{2}[(\cot \alpha + \cot \beta)n^2 - 2tn \cot \beta]} \quad (46)$$

By taking FRFT on both sides of above equation, we have

$$F_{\phi}^{\beta}(u) = F_{\phi}^{\beta}(u) \sum_{n \in \mathbb{Z}} c_n e^{\frac{i}{2}(n^2 \cot \alpha - 2nucsc\beta)}$$

$$\iff F_{\phi}^{\beta}(u/\lambda) = F_{\phi}^{\beta}(u/\lambda) \sum_{n \in \mathbb{Z}} c_n e^{\frac{i}{2}(n^2 \cot \alpha - 2nucsc\alpha)}$$

$$\iff F_{\phi}^{\beta}(u/\lambda) = F_{\phi}^{\beta}(u/\lambda) \tilde{F}_{c'_n}(ucsc\alpha)$$

where $c'_n = c_n e^{\frac{i}{2}n^2 \cot \alpha}$. Then, by utilizing Eq. (44), we have

$$\tilde{F}_{c'_n}(ucsc\alpha) = \frac{1}{\sqrt{2\pi} \sin \beta G_{\phi, \alpha, \beta}(u)} \quad (47)$$

Meanwhile, since $\tilde{F}_{c'_n}(ucsc\alpha)$ is $2\pi \sin \alpha$ periodic, then

$$|F_{\phi}^{\beta}(\frac{u + 2k\pi \sin \alpha}{\lambda})|^2 = |F_{\phi}^{\beta}(\frac{u + 2k\pi \sin \alpha}{\lambda})|^2 |\tilde{F}_{c'_n}(ucsc\alpha)|^2 \quad (48)$$

Taking a sum for all k on both sides of Eq. (48), then

$$G_{\phi, \alpha, \beta}^2(u) = |\tilde{F}_{c'_n}(ucsc\alpha)|^2 G_{\phi, \alpha, \beta}^2(u) \quad (49)$$

Inserting Eq. (49) into Eq. (47) yields

$$G_{\phi, \alpha, \beta}^2(u) = \frac{1}{2\pi \sin \beta} \quad (50)$$

Therefore, the set of functions $\{\phi_{\alpha, \beta, 0, n} = \phi(t - n) \times e^{-\frac{i}{2}[(t^2 - n^2) \cot \alpha - (t - n)^2 \cot \beta]}\}_{n \in \mathbb{Z}}$ forms an orthonormal basis of $V_0^{\alpha, \beta}$.

4.2. Construction of orthogonal fractional wavelets

The orthogonal fractional wavelets can be constructed based on the developed MRA associated with FRWT. The subspace $W_k^{\alpha, \beta}$ is defined as the orthogonal complement of $V_k^{\alpha, \beta}$ in $V_{k+1}^{\alpha, \beta}$, i.e.,

$$W_k^{\alpha, \beta} \perp V_k^{\alpha, \beta}, \quad V_{k+1}^{\alpha, \beta} = W_k^{\alpha, \beta} \oplus V_k^{\alpha, \beta}, \quad \forall k \in \mathbb{Z} \quad (51)$$

Then, according to Definition 4, we can obtain that $\{W_k^{\alpha, \beta}\}_{k \in \mathbb{Z}}$ possess the following properties:

- (1) $W_k^{\alpha, \beta} \perp V_l^{\alpha, \beta}, \forall k \neq l$;
- (2) $\bigoplus_{k \in \mathbb{Z}} W_k^{\alpha, \beta} = L^2(\mathbb{R})$;
- (3) $g(t) \in W_k^{\alpha, \beta} \iff g(2t)e^{\frac{i}{2}[(2t)^2 - t^2] \cot \alpha} \in W_{k+1}^{\alpha, \beta}, \forall k \in \mathbb{Z}$.

The property (2) means that an orthonormal basis of $L^2(\mathbb{R})$ can be constructed by finding out an orthonormal basis of the subspace $W_k^{\alpha, \beta}$. The property (3) implies that a MRA of $L^2(\mathbb{R})$ related to FRWT can be changed to construct the orthonormal basis of $W_0^{\alpha, \beta}$. Therefore, the crucial point is to construct a function $\psi(t) \in L^2(\mathbb{R})$ such that the set of functions $\{\psi_{\alpha, \beta, 0, n} = \psi(t - n)e^{-\frac{i}{2}[(t^2 - n^2) \cot \alpha - (t - n)^2 \cot \beta]}\}_{n \in \mathbb{Z}}$ forms an orthonormal basis of $W_0^{\alpha, \beta}$.

Since $\{\phi_{\alpha, \beta, 1, n} = 2^{\frac{1}{2}}\phi(2t - n)e^{-\frac{i}{2}[(t^2 - (\frac{n}{2})^2] \cot \alpha}\}_{n \in \mathbb{Z}}$ form an orthonormal basis of the subspace $V_1^{\alpha, \beta}$ and $\phi_{\alpha, \beta, 0, 0} = \phi(t) \times e^{-\frac{i}{2}(\cot \alpha - \cot \beta)t^2} \in V_0^{\alpha, \beta} \subseteq V_1^{\alpha, \beta}$, there exist $\{h_n\}_{n \in \mathbb{Z}} \in \ell^2(\mathbb{Z})$ such that

$$\phi_{\alpha, \beta, 0, 0} = \phi(t)e^{-\frac{i}{2}(\cot \alpha - \cot \beta)t^2} = \sum_{n \in \mathbb{Z}} h_n \phi_{\alpha, \beta, 1, n}(t)$$

$$= \sum_{n \in \mathbb{Z}} h_n \sqrt{2}\phi(2t - n)e^{-\frac{i}{2}(t^2 - (\frac{n}{2})^2) \cot \alpha + \frac{i}{2}(2t - n)^2 \cot \beta}$$

Then,

$$\phi(t) = \sum_{n \in \mathbb{Z}} h_n \sqrt{2}\phi(2t - n)e^{\frac{i}{2}[\frac{n^2}{4} \cot \alpha + (3t^2 - 4tn + n^2) \cot \beta]} \quad (52)$$

where $h_n = \sqrt{2}e^{-\frac{i}{2}(\frac{n^2}{4} \cot \alpha + n^2 \cot \beta)} \int_{\mathbb{R}} \phi(t)\overline{\phi(2t - n)} \times e^{-\frac{i}{2}(3t^2 - 4tn) \cot \beta} dt$. Taking FRFT on both sides of Eq. (52), then

$$F_{\phi}^{\beta}(u) = \mathcal{F}^{\beta}[\sum_{n \in \mathbb{Z}} h_n \sqrt{2}\phi(2t - n)e^{\frac{i}{2}[\frac{n^2}{4} \cot \alpha + (3t^2 - 4tn + n^2) \cot \beta]}](u)$$

$$= \frac{\sqrt{2}}{2} e^{\frac{i}{2}\frac{3u^2}{4} \cot \beta} \sum_{n \in \mathbb{Z}} h_n e^{\frac{i}{2}(\frac{n^2}{4} \cot \alpha - nucsc\beta)} \cdot F_{\phi}^{\beta}(\frac{u}{2}) \quad (53)$$

and thus

$$F_{\phi}^{\beta}(u/\lambda) = e^{\frac{i}{2}\frac{3}{4}(\frac{u}{\lambda})^2 \cot \beta} \Lambda(\frac{u}{2\lambda}) \cdot F_{\phi}^{\beta}(\frac{u}{2\lambda}) \quad (54)$$

where

$$\Lambda(u/\lambda) = \frac{\sqrt{2}}{2} \sum_{n \in \mathbb{Z}} h_n e^{\frac{i}{2}(\frac{n^2}{4} \cot \alpha - nucsc\alpha)}. \quad (55)$$

It can be verified that $\Lambda(u/\lambda)$ is a $2\pi \sin \alpha$ period function. Since $\{\phi_{\alpha, \beta, 0, n} = \phi(t - n)e^{-\frac{i}{2}[(t^2 - n^2) \cot \alpha - (t - n)^2 \cot \beta]}\}$ is an orthonormal basis of the subspace $V_0^{\alpha, \beta}$, we have

$$\sum_{k \in \mathbb{Z}} |F_{\phi}^{\beta}(\frac{u + 2k\pi \sin \alpha}{\lambda})|^2 = \frac{1}{2\pi \sin \beta} \quad (56)$$

Moreover,

$$\sum_{k \in \mathbb{Z}} |F_{\phi}^{\beta}(\frac{u + 2k\pi \sin \alpha}{\lambda})|^2$$

$$= \sum_{k \in \mathbb{Z}} |\Lambda(\frac{u + 2k\pi \sin \alpha}{2\lambda}) \cdot F_{\phi}^{\beta}(\frac{u + 2k\pi \sin \alpha}{2\lambda})|^2$$

$$= \sum_{l \in \mathbb{Z}} |\Lambda(\frac{u + 4l\pi \sin \alpha}{2\lambda}) F_{\phi}^{\beta}(\frac{u + 4l\pi \sin \alpha}{2\lambda})|^2$$

$$+ \sum_{l \in \mathbb{Z}} |\Lambda(\frac{u + (4l + 2)\pi \sin \alpha}{2\lambda}) F_{\phi}^{\beta}(\frac{u + (4l + 2)\pi \sin \alpha}{2\lambda})|^2$$

$$= \sum_{l \in \mathbb{Z}} |\Lambda(\frac{u/2 + 2l\pi \sin \alpha}{\lambda}) F_{\phi}^{\beta}(\frac{u/2 + 2l\pi \sin \alpha}{\lambda})|^2$$

$$+ \sum_{l \in \mathbb{Z}} |\Lambda(\frac{u/2 + (2l + 1)\pi \sin \alpha}{\lambda}) F_{\phi}^{\beta}(\frac{u/2 + (2l + 1)\pi \sin \alpha}{\lambda})|^2$$

$$= |\Lambda(\frac{u}{2\lambda})|^2 \sum_{l \in \mathbb{Z}} |F_{\phi}^{\beta}(\frac{u/2 + 2l\pi \sin \alpha}{\lambda})|^2$$

$$+ |\Lambda(\frac{u + 2\pi \sin \alpha}{2\lambda})|^2 \sum_{l \in \mathbb{Z}} |F_{\phi}^{\beta}(\frac{u/2 + (2l + 1)\pi \sin \alpha}{\lambda})|^2$$

$$= |\Lambda(\frac{u}{2\lambda})|^2 + |\Lambda(\frac{u + 2\pi \sin \alpha}{2\lambda})|^2$$

It can be deduced that

$$|\Lambda(\frac{u}{\lambda})|^2 + |\Lambda(\frac{u + \pi \sin \alpha}{\lambda})|^2 = 1 \quad (57)$$

Since $\psi_{\alpha, \beta, 0, 0} = \psi(t)e^{-\frac{i}{2}[(\cot \alpha - \cot \beta)t^2]} \in W_0^{\alpha, \beta} \subseteq V_1^{\alpha, \beta}$, there exist $\{g_n\}_{n \in \mathbb{Z}} \in \ell^2(\mathbb{Z})$ such that

$$\psi_{\alpha, \beta, 0, 0} = \psi(t)e^{-\frac{i}{2}[(\cot \alpha - \cot \beta)t^2]} = \sum_{n \in \mathbb{Z}} g_n \phi_{\alpha, \beta, 1, n}(t) \quad (58)$$

Taking FRFT on both sides of above equation, then

$$F_{\psi}^{\beta}(u/\lambda) = e^{\frac{j}{2}\frac{u}{\lambda} \cot \beta} \Gamma\left(\frac{u}{2\lambda}\right) \cdot F_{\phi}^{\beta}\left(\frac{u}{2\lambda}\right), \quad (59)$$

where

$$\Gamma(u/\lambda) = \frac{\sqrt{2}}{2} \sum_{n \in \mathbb{Z}} g_n e^{\frac{j}{2}(\frac{n^2}{4} \cot \alpha - 2nucsc\alpha)}. \quad (60)$$

Then, we have

$$|\Gamma(\frac{u}{\lambda})|^2 + |\Gamma(\frac{u + \pi \sin \alpha}{\lambda})|^2 = 1 \quad (61)$$

Moreover, since $W_0^{\alpha, \beta}$ and $V_0^{\alpha, \beta}$ are orthogonal in $V_1^{\alpha, \beta}$, then

$$\langle \phi_{\alpha, \beta, 0, m}(t), \psi_{\alpha, \beta, 0, n}(t) \rangle_{L^2} = 0, \quad \forall m, n \in \mathbb{Z} \quad (62)$$

And it can be easily deduced the following two results based on the similar derivation of Eq. (38)

$$\mathcal{F}^{\alpha}[\phi_{\alpha, \beta, 0, m}(t)](u) = \frac{1}{A_{\beta}} e^{-\frac{j}{2}(\frac{u}{\lambda})^2 \cot \beta} K_{\alpha}(u, m) F_{\phi}^{\beta}(u/\lambda) \quad (63)$$

$$\mathcal{F}^{\alpha}[\psi_{\alpha, \beta, 0, n}(t)](u) = \frac{1}{A_{\beta}} e^{-\frac{j}{2}(\frac{u}{\lambda})^2 \cot \beta} K_{\alpha}(u, n) F_{\psi}^{\beta}(u/\lambda) \quad (64)$$

By using Eqs. (9), (63) and (64), we have

$$\begin{aligned} & \langle \phi_{\alpha, \beta, 0, m}(t), \psi_{\alpha, \beta, 0, n}(t) \rangle_{L^2} \\ &= \langle \mathcal{F}^{\alpha}[\phi_{\alpha, \beta, 0, m}(t)](u), \mathcal{F}^{\alpha}[\psi_{\alpha, \beta, 0, n}(t)](u) \rangle_{L^2} \\ &= \frac{1}{A_{\beta}^2} \int_{\mathbb{R}} K_{\alpha}(u, m) F_{\phi}^{\beta}(u/\lambda) \cdot \overline{K_{\alpha}(u, n) F_{\psi}^{\beta}(u/\lambda)} du \\ &= \frac{1}{A_{\beta}^2} \int_{\mathbb{R}} \Lambda\left(\frac{u}{2\lambda}\right) \overline{\Gamma\left(\frac{u}{2\lambda}\right)} |F_{\phi}^{\beta}\left(\frac{u}{2\lambda}\right)|^2 K_{\alpha}(u, m) \overline{K_{\alpha}(u, n)} du \\ &= \frac{1}{A_{\beta}^2} \sum_{k \in \mathbb{Z}} \int_0^{4\pi \sin \alpha} \Lambda\left(\frac{u + 4k\pi \sin \alpha}{2\lambda}\right) \\ & \quad \cdot \overline{\Gamma\left(\frac{u + 4k\pi \sin \alpha}{2\lambda}\right)} |F_{\phi}^{\beta}\left(\frac{u + 4k\pi \sin \alpha}{2\lambda}\right)|^2 \\ & \quad \cdot K_{\alpha}(u + 4k\pi \sin \alpha, m) \overline{K_{\alpha}(u + 4k\pi \sin \alpha, n)} du \\ &= \frac{1}{A_{\beta}^2} \int_0^{4\pi \sin \alpha} \Lambda\left(\frac{u}{2\lambda}\right) \overline{\Gamma\left(\frac{u}{2\lambda}\right)} \sum_{k \in \mathbb{Z}} |F_{\phi}^{\beta}\left(\frac{u + 4k\pi \sin \alpha}{2\lambda}\right)|^2 \\ & \quad \cdot K_{\alpha}(u, m) \overline{K_{\alpha}(u, n)} du \\ &= \frac{1}{A_{\beta}^2} \cdot \frac{1}{2\pi \sin \beta} \int_0^{2\pi \sin \alpha} \Lambda\left(\frac{u}{2\lambda}\right) \overline{\Gamma\left(\frac{u}{2\lambda}\right)} K_{\alpha}(u, m) \overline{K_{\alpha}(u, n)} du \\ & \quad + \frac{1}{A_{\beta}^2} \cdot \frac{1}{2\pi \sin \beta} \int_{2\pi \sin \alpha}^{4\pi \sin \alpha} \Lambda\left(\frac{u}{2\lambda}\right) \overline{\Gamma\left(\frac{u}{2\lambda}\right)} K_{\alpha}(u, m) \overline{K_{\alpha}(u, n)} du \\ &= \frac{A_{\alpha}^2}{A_{\beta}^2} \cdot \frac{1}{2\pi \sin \beta} e^{\frac{j}{2}(m^2 - n^2) \cot \alpha} \\ & \quad \cdot \int_0^{2\pi \sin \alpha} \left[\Lambda\left(\frac{u}{2\lambda}\right) \overline{\Gamma\left(\frac{u}{2\lambda}\right)} + \Lambda\left(\frac{u + 2\pi \sin \alpha}{2\lambda}\right) \overline{\Gamma\left(\frac{u + 2\pi \sin \alpha}{2\lambda}\right)} \right] \\ & \quad \cdot e^{-i(m-n)csc\alpha} du \end{aligned} \quad (65)$$

Because $\{\frac{1}{\sqrt{2\pi}} e^{-inucsc\alpha}\}_{n \in \mathbb{Z}}$ is a orthonormal basis of $L^2[0, 2\pi \sin \alpha]$, we obtain

$$\Lambda\left(\frac{u}{\lambda}\right) \overline{\Gamma\left(\frac{u}{\lambda}\right)} + \Lambda\left(\frac{u + \pi \sin \alpha}{\lambda}\right) \overline{\Gamma\left(\frac{u + \pi \sin \alpha}{\lambda}\right)} = 0 \quad a.e. \quad \forall u \in \mathbb{R} \quad (66)$$

Overall, the following theorem has been proved.

Theorem 4. The set of functions $\{\psi_{\alpha, \beta, 0, n} = 2^{\frac{1}{2}} \psi(t - n) \times e^{-\frac{j}{2}[(t^2 - n^2) \cot \alpha - (t - n)^2 \cot \beta]}\}_{n \in \mathbb{Z}}$ is an orthogonal basis of $W_0^{\alpha, \beta}$ if and only if $\mathbf{M}(u/\lambda)$ is a unitary matrix, i.e.,

$$\mathbf{M}(u/\lambda) \overline{\mathbf{M}(u/\lambda)} = \mathbf{I}, \quad a.e. \quad \forall u \in \mathbb{R} \quad (67)$$

where

$$\mathbf{M}(u/\lambda) = \begin{pmatrix} \Lambda\left(\frac{u}{\lambda}\right) & \Lambda\left(\frac{u + \pi \sin \alpha}{\lambda}\right) \\ \Gamma\left(\frac{u}{\lambda}\right) & \Gamma\left(\frac{u + \pi \sin \alpha}{\lambda}\right) \end{pmatrix} \quad (68)$$

If the Eq. (67) holds, then there exists a function $\rho(u/\lambda)$ such that

$$\left(\Gamma\left(\frac{u}{\lambda}\right), \Gamma\left(\frac{u + \pi \sin \alpha}{\lambda}\right)\right) = \left(\rho\left(\frac{u}{\lambda}\right) \overline{\Lambda\left(\frac{u + \pi \sin \alpha}{\lambda}\right)}, -\rho\left(\frac{u}{\lambda}\right) \overline{\Lambda\left(\frac{u}{\lambda}\right)}\right) \quad (69)$$

Since $\Gamma(\frac{u}{\lambda})$ and $\overline{\Lambda(\frac{u + \pi \sin \alpha}{\lambda})}$ are $2\pi \sin \alpha$ periodic function, $\rho(\frac{u}{\lambda})$ is a $2\pi \sin \alpha$ periodic function. Therefore, $\rho(u/\lambda)$ can be expanded as a Fourier series, i.e.,

$$\rho(u/\lambda) = \sum_{k \in \mathbb{Z}} c_k e^{-ikucsc\alpha}, \quad (70)$$

where

$$\begin{aligned} c_k &= \frac{1}{2\pi \sin \alpha} \int_0^{2\pi \sin \alpha} \rho\left(\frac{u}{\lambda}\right) e^{ikucsc\alpha} du \\ &= \frac{1}{2\pi \sin \alpha} \int_0^{\pi \sin \alpha} \rho\left(\frac{u}{\lambda}\right) e^{ikucsc\alpha} du + \int_{\pi \sin \alpha}^{2\pi \sin \alpha} \rho\left(\frac{u}{\lambda}\right) e^{ikucsc\alpha} du \\ &= \frac{1}{2\pi \sin \alpha} \int_0^{\pi \sin \alpha} \rho\left(\frac{u}{\lambda}\right) e^{ikucsc\alpha} du \\ & \quad + \int_0^{\pi \sin \alpha} \rho\left(\frac{u + \pi \sin \alpha}{\lambda}\right) e^{ik(u + \pi \sin \alpha)csc\alpha} du \\ &= \frac{1 - (-1)^k}{2\pi \sin \alpha} \int_0^{\pi \sin \alpha} \rho\left(\frac{u}{\lambda}\right) e^{ikucsc\alpha} du \end{aligned} \quad (71)$$

$\rho(u/\lambda)$ can be rewritten as

$$\begin{aligned} \rho\left(\frac{u}{\lambda}\right) &= \sum_{l \in \mathbb{Z}} c_{2l+1} e^{-i(2l+1)ucsc\alpha} = e^{-iucsc\alpha} \sum_{l \in \mathbb{Z}} c_{2l+1} e^{-i2lucsc\alpha} \\ &= e^{-iucsc\alpha} \gamma\left(\frac{2u}{\lambda}\right) \end{aligned} \quad (72)$$

where $\gamma(u/\lambda) = \sum_{l \in \mathbb{Z}} c_{2l+1} e^{-ilucsc\alpha}$.

Moreover, followed by the Eq. (55) and Eq. (60), we have

$$\begin{aligned}\Lambda\left(\frac{u+\pi\sin\alpha}{\lambda}\right) &= \frac{\sqrt{2}}{2} \sum_{n \in \mathbb{Z}} \overline{h_n} e^{-\frac{i}{2}(\frac{n^2}{4} \cot\alpha - 2n(u+\pi\sin\alpha)\csc\alpha)} \\ &= \frac{\sqrt{2}}{2} \sum_{n \in \mathbb{Z}} (-1)^n \overline{h_n} e^{-\frac{i}{2}(\frac{n^2}{4} \cot\alpha - 2nucsc\alpha)}\end{aligned}\quad (73)$$

$$\Gamma(u/\lambda) = \frac{\sqrt{2}}{2} \sum_{n \in \mathbb{Z}} g_n e^{\frac{i}{2}(\frac{n^2}{4} \cot\alpha - 2nucsc\alpha)}.\quad (74)$$

Thus, followed by the Eq. (69), then

$$\Gamma\left(\frac{u}{\lambda}\right) = e^{-iucsc\alpha} \gamma\left(\frac{2u}{\lambda}\right) \Lambda\left(\frac{u+\pi\sin\alpha}{\lambda}\right)\quad (75)$$

Moreover, let $\gamma\left(\frac{2u}{\lambda}\right) = 1$, we have

$$\begin{aligned}& g_k e^{\frac{i}{2} \frac{k^2}{4} \cot\alpha} \\ &= \int_{\mathbb{R}} e^{-iucsc\alpha} \cdot \sum_{n \in \mathbb{Z}} (-1)^n \overline{h_n} e^{-\frac{i}{2}(\frac{n^2}{4} \cot\alpha - 2nucsc\alpha)} \cdot e^{ikucsc\alpha} du \\ &= \sum_{n \in \mathbb{Z}} (-1)^n \overline{h_n} e^{-\frac{i}{2} \frac{n^2}{4} \cot\alpha} \int_{\mathbb{R}} e^{i(k+n-1)ucsc\alpha} du \\ &= (-1)^{1-k} \overline{h_{1-k}} e^{-\frac{i}{2} \frac{(1-k)^2}{4} \cot\alpha}\end{aligned}\quad (76)$$

An example for constructing orthogonal fractional Meyer wavelet is shown. The scale function $\phi(t)$ is defined by

$$F_{\phi}^{\beta}\left(\frac{u}{\lambda}\right) = \begin{cases} (2\pi \sin \beta)^{-1/2} & |u| \leq u_{\alpha} \\ (0, (2\pi \sin \beta)^{-1/2}) & u_{\alpha} < |u| < 2u_{\alpha} \\ 0 & |u| \geq 2u_{\alpha} \end{cases}\quad (77)$$

where $u_{\alpha} = \frac{2\pi|\sin\alpha|}{3}$. It can be verified that $G_{\phi, \alpha, \beta}^2 = \sum_{k \in \mathbb{Z}} |F_{\phi}^{\beta}(\frac{u+2k\pi\sin\alpha}{\lambda})|^2 = \frac{1}{2\pi\sin\beta}$, which means that $\{\phi_{\alpha, \beta, 0, n} = \phi(t-n)e^{-\frac{i}{2}[(t^2-n^2)\cot\alpha - (t-n)^2\cot\beta]}\}$ forms an orthonormal basis of the subspace $V_0^{\alpha, \beta}$. Hence, it can be deduced by Eq. (54) that

$$\begin{aligned}\Lambda\left(\frac{u}{\lambda}\right) &= e^{-\frac{i}{2} \frac{3}{4} (\frac{2u}{\lambda})^2 \cot\beta} \frac{F_{\phi}^{\beta}(\frac{2u}{\lambda})}{F_{\phi}^{\beta}(\frac{u}{\lambda})} \\ &= (2\pi \sin \beta)^{1/2} e^{-\frac{i}{2} \frac{3}{4} (\frac{2u}{\lambda})^2 \cot\beta} F_{\phi}^{\beta}\left(\frac{2u}{\lambda}\right)\end{aligned}\quad (78)$$

Then, combined with Eqs. (59) and (75), we have

$$\begin{aligned}& F_{\psi}^{\beta}(u/\lambda) \\ &= e^{\frac{i}{2} \frac{3}{4} (\frac{u}{\lambda})^2 \cot\beta} \Gamma\left(\frac{u}{2\lambda}\right) \cdot F_{\phi}^{\beta}\left(\frac{u}{2\lambda}\right) \\ &= e^{\frac{i}{2} \frac{3}{4} (\frac{u}{\lambda})^2 \cot\beta} e^{-\frac{iucsc\alpha}{2}} \Lambda\left(\frac{u+2\pi\sin\alpha}{2\lambda}\right) \cdot F_{\phi}^{\beta}\left(\frac{u}{2\lambda}\right) \\ &= e^{\frac{i}{2} \frac{3}{4} (\frac{u}{\lambda})^2 \cot\beta} e^{-\frac{iucsc\alpha}{2}} (2\pi \sin \beta)^{1/2} e^{\frac{i}{2} \frac{3}{4} (\frac{u+2\pi\sin\alpha}{\lambda})^2 \cot\beta} \\ &\quad \cdot F_{\phi}^{\beta}\left(\frac{u+2\pi\sin\alpha}{\lambda}\right) \cdot F_{\phi}^{\beta}\left(\frac{u}{2\lambda}\right)\end{aligned}$$

Then, $\psi(t)$ can be obtained by applying the inverse FRFT on $F_{\psi}^{\beta}(u/\lambda)$.

5. Application

5.1. Time-fractional-frequency analysis

According to Eq. (14), $W_f^{\alpha, \beta}(a, b)$ is the inner product of $f(t)$ and $\psi_{a, b, \alpha, \beta}(t)$. It means that $W_f^{\alpha, \beta}(a, b)$ is supported in the time domain if the kernel $\psi_{a, b, \alpha, \beta}(t)$ is supported in the time domain. Similarly, the multiplication of $F^{\alpha}(u)$ and $F_{\psi}^{\beta}(au/\lambda)$ can provide local properties of $f(t)$ in the fractional domain if $F_{\psi}^{\beta}(\Omega)$ satisfies the admissibility condition (see Eq. (24)). This implies that FRWT can provide the information of signal in time domain and fractional domain at the same time. That is to say, FRWT is capable of giving a time-fractional-frequency representation for signal. In this section, the constant-Q property and TFF resolution of FRWT are deduced firstly. Then, LFM signal is selected as an example to show the better performance of FRWT in signal representation.

5.1.1. Constant-Q property

Suppose that $\psi(t)$ is a function with finite center $E\psi$ and radii $\Delta\psi$ in the time domain, F_{ψ}^{β} is a function with finite center $E[F_{\psi}^{\beta}]$ and radii ΔF_{ψ}^{β} in the fractional domain with parameter β . Then, the center and radius of window function $\psi_{a, b, \alpha, \beta}(t)$ in the time domain are respectively given by

$$\begin{aligned}E[\psi_{a, b, \alpha, \beta}(t)] &= \frac{\int_{\mathbb{R}} t |\psi_{a, b, \alpha, \beta}(t)|^2 dt}{\|\psi_{a, b, \alpha, \beta}(t)\|^2} = \frac{\int_{\mathbb{R}} t |\psi_{a, b}(t)|^2 dt}{\|\psi_{a, b}(t)\|^2} \\ &= E[\psi_{a, b}(t)] = b + aE\psi\end{aligned}\quad (79)$$

$$\begin{aligned}\Delta[\psi_{a, b, \alpha, \beta}(t)] &= \left(\frac{\int_{\mathbb{R}} (t - E[\psi_{a, b, \alpha, \beta}(t)])^2 |\psi_{a, b, \alpha, \beta}(t)|^2 dt}{\|\psi_{a, b, \alpha, \beta}(t)\|^2} \right)^{1/2} \\ &= \left(\frac{\int_{\mathbb{R}} (t - E[\psi_{a, b}(t)])^2 |\psi_{a, b}(t)|^2 dt}{\|\psi_{a, b}(t)\|^2} \right)^{1/2} \\ &= \Delta[\psi_{a, b}(t)] = a\Delta\psi\end{aligned}\quad (80)$$

where E_{ψ}, Δ_{ψ} denotes the finite center and radius of $\psi(t)$ in the time domain.

Then, the center and radius of window function $F_{\psi_{a, b, \alpha, \beta}}^{\alpha}(u)$ in the fractional domain are respectively given by

$$\begin{aligned}E[F_{\psi_{a, b, \alpha, \beta}}^{\alpha}(u)] &= \frac{\int_{\mathbb{R}} u |F_{\psi_{a, b, \alpha, \beta}}^{\alpha}(u)|^2 du}{\|F_{\psi_{a, b, \alpha, \beta}}^{\alpha}(u)\|^2} = \frac{\int_{\mathbb{R}} u |F_{\psi}^{\beta}(au/\lambda)|^2 du}{\|F_{\psi}^{\beta}(au/\lambda)\|^2} \\ &= \frac{\int_{\mathbb{R}} \xi |F_{\psi}^{\beta}(\xi)|^2 d\xi}{\|F_{\psi}^{\beta}(\xi)\|^2} \quad (\text{let } \xi = au/\lambda) \\ &= \frac{\lambda}{a} E[F_{\psi}^{\beta}]\end{aligned}\quad (81)$$

$$\begin{aligned}
& \Delta[F_{\psi_{a,b,\alpha,\beta}}^\alpha(u)] \\
&= \left(\frac{\int_{\mathbb{R}} (u - E[F_{\psi_{a,b,\alpha,\beta}}^\alpha(u)])^2 |F_{\psi_{a,b,\alpha,\beta}}^\alpha(u)|^2 du}{\|F_{\psi_{a,b,\alpha,\beta}}^\alpha(u)\|^2} \right)^{1/2} \\
&= \left(\frac{\int_{\mathbb{R}} (u - \frac{\lambda}{a} E[F_{\psi}^\beta(au/\lambda)])^2 |F_{\psi}^\beta(au/\lambda)|^2 \frac{a}{\lambda} du}{\|F_{\psi}^\beta(au/\lambda)\|^2} \right)^{1/2} \\
&= \left(\frac{\int_{\mathbb{R}} (\frac{\eta\lambda}{a} - \frac{\lambda}{a} E[F_{\psi}^\beta(\eta)])^2 |F_{\psi}^\beta(\eta)|^2 d\eta}{\|F_{\psi}^\beta(\eta)\|^2} \right)^{1/2} \quad (\text{let } \eta = au/\lambda) \\
&= \frac{\lambda}{a} \Delta F_{\psi}^\beta
\end{aligned} \tag{82}$$

Therefore, the Q-factor of fractional domain window function $F_{\psi_{a,b,\alpha,\beta}}^\alpha(u)$ is given by

$$Q = \frac{\Delta[F_{\psi_{a,b,\alpha,\beta}}^\alpha(u)]}{E[F_{\psi_{a,b,\alpha,\beta}}^\alpha(u)]} = \frac{\frac{\lambda}{a} \Delta F_{\psi}^\beta}{\frac{\lambda}{a} E[F_{\psi}^\beta]} = \frac{\Delta F_{\psi}^\beta}{E[F_{\psi}^\beta]} \tag{83}$$

From Eq. (83), the Q-factor of $F_{\psi_{a,b,\alpha,\beta}}^\alpha(u)$ is independent of the transform parameter α and scaling parameter a . It means that Q-factor is constant when β is fixed, this is the constant-Q property of proposed FRWT.

5.1.2. Time-fractional-frequency resolution

From Eqs. (14), (79) and (80), it can be seen that the FRWT of $f(t)$ localizes the signal with a window in the time domain. The time window is

$$[b + aE\psi - a\Delta\psi, b + aE\psi + a\Delta\psi] \tag{84}$$

Moreover, from Eqs. (21), (81) and (82), the FRWT of $f(t)$ also localizes the signal with a window in the fractional domain. The fractional window is

$$\left[\frac{\lambda}{a} E[F_{\psi}^\beta] - \frac{\lambda}{a} \Delta F_{\psi}^\beta, \frac{\lambda}{a} E[F_{\psi}^\beta] + \frac{\lambda}{a} \Delta F_{\psi}^\beta \right] \tag{85}$$

Therefore, according to the definition of FRWT in the time domain (see Eq. (14)) and fractional domain (see Eq. (21)), the FRWT of a signal provides the localization analysis in the time and fractional domain, determines a time-fractional-frequency analysis window as follows

$$\begin{aligned}
& [b + aE\psi - a\Delta\psi, b + aE\psi + a\Delta\psi] \\
& \times \left[\frac{\lambda}{a} E[F_{\psi}^\beta] - \frac{\lambda}{a} \Delta F_{\psi}^\beta, \frac{\lambda}{a} E[F_{\psi}^\beta] + \frac{\lambda}{a} \Delta F_{\psi}^\beta \right]
\end{aligned} \tag{86}$$

The area of this analysis window in the time-fractional-frequency plane is

$$2a\Delta\psi \times 2\frac{\lambda}{a} \Delta F_{\psi}^\beta = 4\lambda\Delta\psi \Delta F_{\psi}^\beta \tag{87}$$

The area only depends on the mother wavelet $\psi(t)$ and the parameters α, β . For a fixed α, β , the shape of time-fractional-frequency window varies with the scaling parameter a . According to Eq. (87), it can

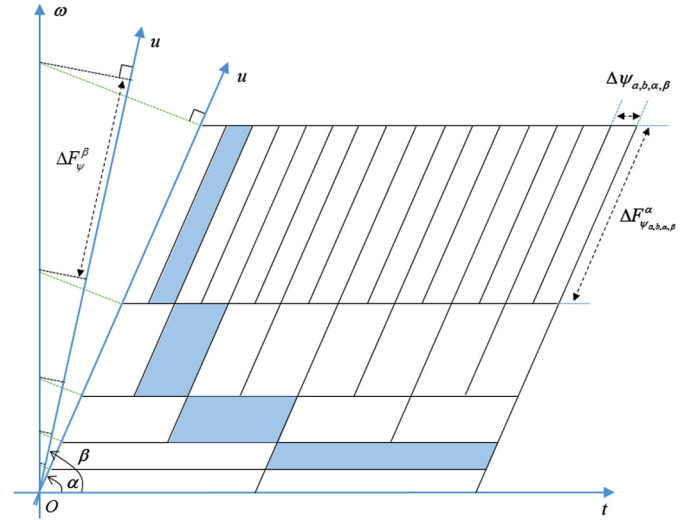


Fig. 1. Time-fractional-frequency analysis windows of FRWT.

be seen that how the time and fractional resolutions change with the width and height of analysis window. The analysis window of FRWT in the time-fractional-frequency plane is shown in Fig. 1.

5.2. LFM signal TFF analysis

5.2.1. LFM signal

A single component LFM signal $f(t)$ can be expressed as

$$f(t) = \text{rect}(t/T) \cdot A e^{\frac{j}{2}(kt^2 + 2f_0 t)} \tag{88}$$

where A, T, k, f_0 denotes the amplitude, time width, chirp rate, center frequency respectively, $\text{rect}(\cdot)$ is rectangular function. The instantaneous frequency of LFM signal is $IF(t) = kt + f_0$.

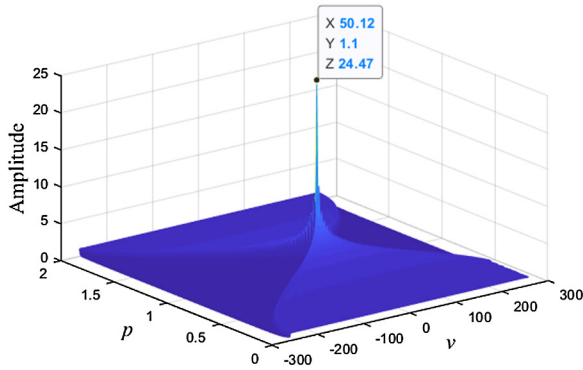
The FRFT amplitude spectrum of LFM signal can be given by following three cases:

- (1) when $\cot\alpha = -k$, the amplitude spectrum of LFM signal with finite length in the optimal FRFT domain obeys sinc function distribution, i.e.,

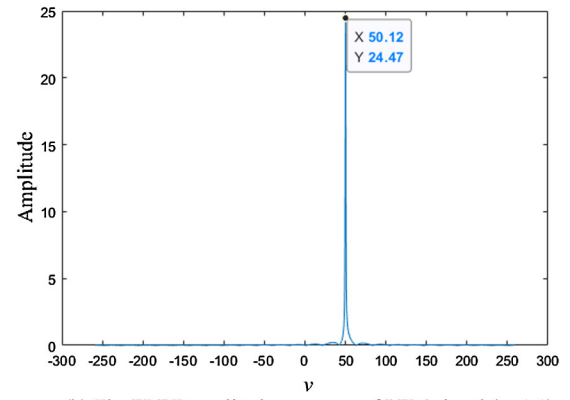
$$F_f^\alpha(u) = \mathcal{F}^\alpha[f(t)](u) = AA_\alpha e^{\frac{j}{2}u^2 \cot\alpha} T \text{sinc}\left(\frac{ucsc\alpha - f_0}{2\pi/T}\right) \tag{89}$$

- (2) when $T\sqrt{\cot\alpha + k} \gg 1$, with the increase of $T\sqrt{\cot\alpha + k}$, the amplitude spectrum of LFM signal tends to be flat in the support region, its shape is close to a rectangle. The signal energy is mainly concentrated in the support region, while the energy is very small outside the support region.
- (3) when $T\sqrt{\cot\alpha + k}$ decreases and approaches to 0 ($T\sqrt{\cot\alpha + k} \rightarrow 0$), the shape of amplitude spectrum changes from rectangle to sinc function, and the amplitude increases rapidly.

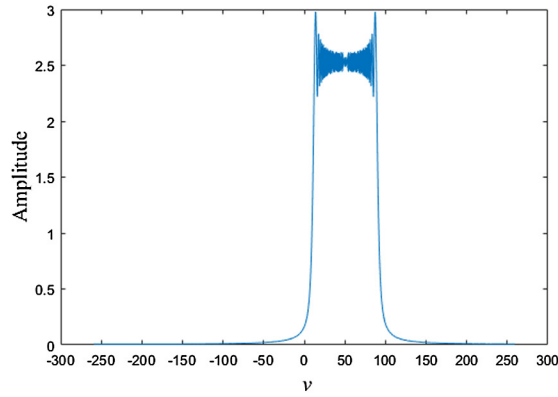
A LFM signal with parameters $\{A, T, k, f_0, f_s\} = \{1, 4, 20, 50, 2^9\}$ is selected to test above results. Let $\alpha = p\pi/2$, p changes from 0.01 to 2 with 0.01 intervals. The FRFT amplitude spectrum with different p are shown in the Fig. 2 (a), where $v = u \csc\alpha$ denotes the fractional frequency with scale $\csc\alpha$. From Fig. 2 (a), the FRFT amplitude spectrum of LFM signal shows an energy concentration in a certain fractional domain (i.e., $p = 1.1$), and the corresponding amplitude reaches the global maximum in the $p-v$ plane. The FRFT amplitude spectrum with $p = 1.1, 1, 1.3$ are plotted in Fig. 2 (b)-(d). For a LFM signal with center frequency f_0 and chirp rate k , the following results can be found from these figures:



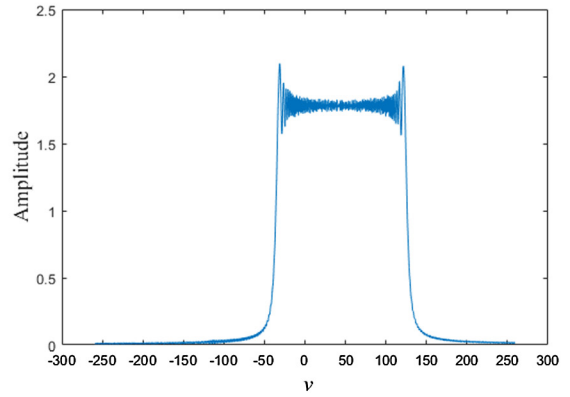
(a) The 3D graph of FRFT amplitude spectrum for LFM signal



(b) The FRFT amplitude spectrum of LFM signal (p=1.1)



(c) The FRFT amplitude spectrum of LFM signal (p=1)



(d) The FRFT amplitude spectrum of LFM signal (p=1.3)

Fig. 2. The FRFT amplitude spectrum of LFM signal.

- (1) when α is matched with chirp rate k , its amplitude has a significant peak value at $v = f_0$.
- (2) when α is not matched with chirp rate k , its spectrum tends to be flat in the support region, the shape of spectrum is close to a rectangle. The closer p to 1.1, the narrower the width of rectangle is. It should be emphasized that LFM signal no longer has impulse characteristics in the frequency domain ($p = 1$).

5.2.2. Chirplet

A mother chirplet with Gaussian envelope can be given by

$$\psi(t) = (\pi f_b)^{-1/2} e^{-\frac{t^2}{f_b}} e^{\frac{i}{2}(2f_c t + ct^2)} \quad (90)$$

The FRFT of $\psi(t)$ can be given by

$$\begin{aligned} F_{\psi}^{\beta}(u) &= \int_{\mathbb{R}} (\pi f_b)^{-1/2} e^{-\frac{t^2}{f_b}} e^{\frac{i}{2}(2f_c t + ct^2)} \cdot A_{\beta} e^{\frac{i}{2}[(t^2+u^2)\cot\beta - 2tucsc\beta]} dt \\ &= A_{\beta} (\pi f_b)^{-1/2} e^{\frac{i}{2}u^2\cot\beta} \int_{\mathbb{R}} e^{-[\frac{1}{f_b} - \frac{i}{2}(c+\cot\beta)]t^2} e^{-i(ucsc\beta - f_c)t} dt \\ &= A_{\beta} (f_b \xi)^{-1/2} e^{\frac{i}{2}u^2\cot\beta} e^{-\frac{\xi(ucsc\beta - f_c)^2}{4|\xi|^2}} \end{aligned} \quad (91)$$

where $\xi = \frac{1}{f_b} - \frac{i}{2}(c + \cot\beta)$. The FRFT amplitude spectrum of chirplet is

$$|F_{\psi}^{\beta}(u)| = |A_{\beta}| (f_b |\xi|)^{-1/2} e^{-\frac{(ucsc\beta - f_c)^2}{4f_b |\xi|^2}}. \quad (92)$$

According to the properties of Gaussian distribution, when $\cot\beta = -c$, the FRFT amplitude spectrum of chirplet reaches peak value at $v = ucsc\alpha = f_c$. At this time, larger bandwidth f_b will lead to more centralized distribution.

Three chirplets are selected to show their properties. The real and imaginary parts of each chirplet are shown in Fig. 3, the corresponding FT and FRFT amplitude spectrum with matched α are shown in Fig. 4. From these figures, it is pretty obvious that chirplet has better compact support in the fractional domain than that of Fourier (or frequency) domain. The influence of each parameter on chirplet is concluded as follows: (1) bandwidth f_b affects the compact support of chirplet in the time and fractional domain. With the increase of bandwidth f_b , the support region of chirplet in the time domain is getting wider. On the contrary, increasing f_b results in a narrower concentration of energy around the center frequency f_c in the matched fractional domain; (2) center frequency f_c affects the oscillation degree of chirplet in the time domain and the position of peak value in the fractional domain; (3) chirp rate c directly determines the optimal angle parameters of FRFT, the FRFT amplitude spectrum of chirplet has best energy concentration when β satisfies $\cot\beta = -c$; (4) when the parameters f_b, f_c, c are fixed, the parameter β directly affects the cut-off frequency and passband width of bandpass filter. The closer β to $-\text{arccot}c$, the narrower the passband width of bandpass filter is.

5.2.3. Time-fractional-frequency representation

The proposed FRWT has two free parameters α, β , chirplet has three parameters f_b, f_c, c , it is necessary to analyze the influence

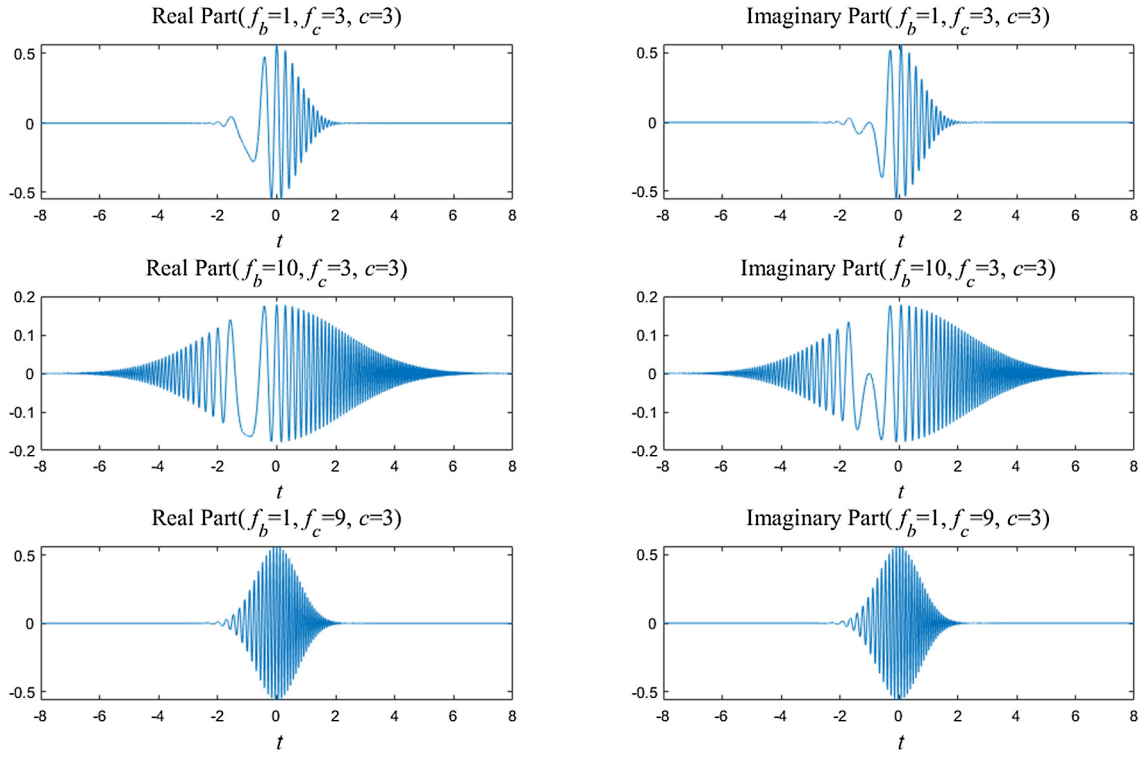


Fig. 3. The chirplet with different parameters.

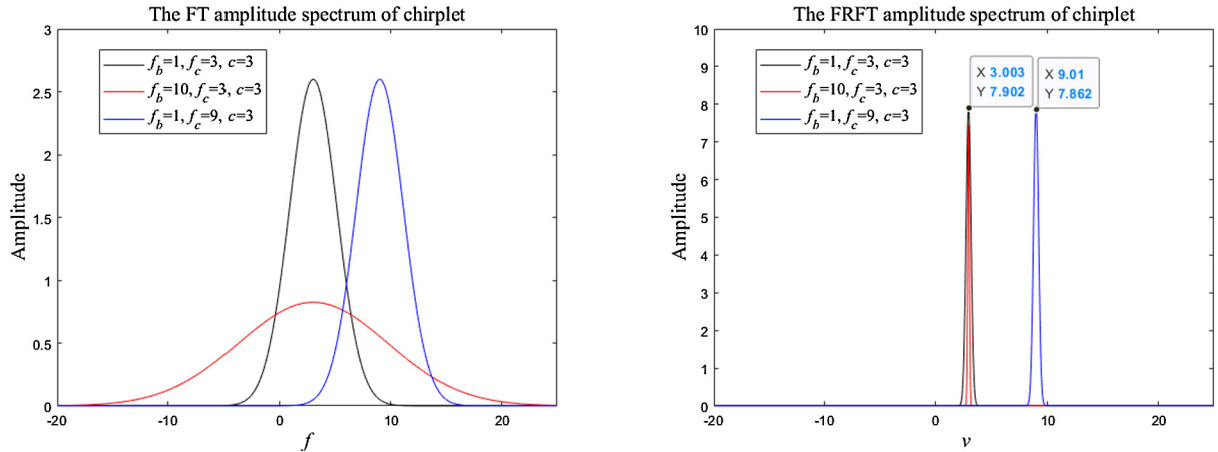


Fig. 4. The FT and FRFT amplitude spectrum of chirplet. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

of these parameters on LFM signal representation. According to the analysis on chirplet and LFM signal, when $\alpha = -\arccot k$, $\beta = -\arccot c$, larger f_b and f_c will lead to better energy concentration. The simulation with different parameters is shown in Fig. 5. Compared with Fig. 6, larger f_b and f_c will make better energy concentration of LFM signal representation, which are consistent with that of theoretical analysis. When α is not the optimal, the slope rotates in the time-frequency plane. To better demonstrate the effect of β on LFM signal representation, the simulation is conducted by three sets of parameters with different β . These results show that the energy concentration of LFM signal representation drops sharply if β has a small error with the optimal angle $-\arccot c$. In brief, the simulation results show that the parameters f_b, f_c, β have great influence on the energy concentration of LFM signal representation.

For a single component LFM signal with parameters $\{A, T, k, f_0, f_s\} = \{1, 4, 20, 50, 2^9\}$, its true instantaneous frequency (IF) is $IF(t) = f_0 + kt$. For pure and noisy single component LFM signal, the simulation experiments are carried out with different methods, the results are shown in Fig. 6 and Fig. 7. The complex Morlet wavelet with parameters $f_b = 10, f_c = 3$ is used in the CWT, chirplet with parameters $f_b = 10, f_c = 3, c = 3$ is used in the FRWT (Dai) and proposed FRWT. From Eqs. (89) and (92), the parameters of FRWT (Dai) are set by $\alpha = \beta = -\arccot k$, the parameters of proposed FRWT are set by $\alpha = -\arccot k, \beta = -\arccot c$. From Fig. 6, the following results can be obtained: 1) Since CT and STFT have fixed time-frequency resolution, WT has low time-frequency resolution in the low frequency part. In comparison with CT, STFT, CWT, the energy concentration of LFM signal representation by proposed FRWT is better than that of traditional methods; 2) For FRWT(Dai), the chirplet has no finite support in fractional

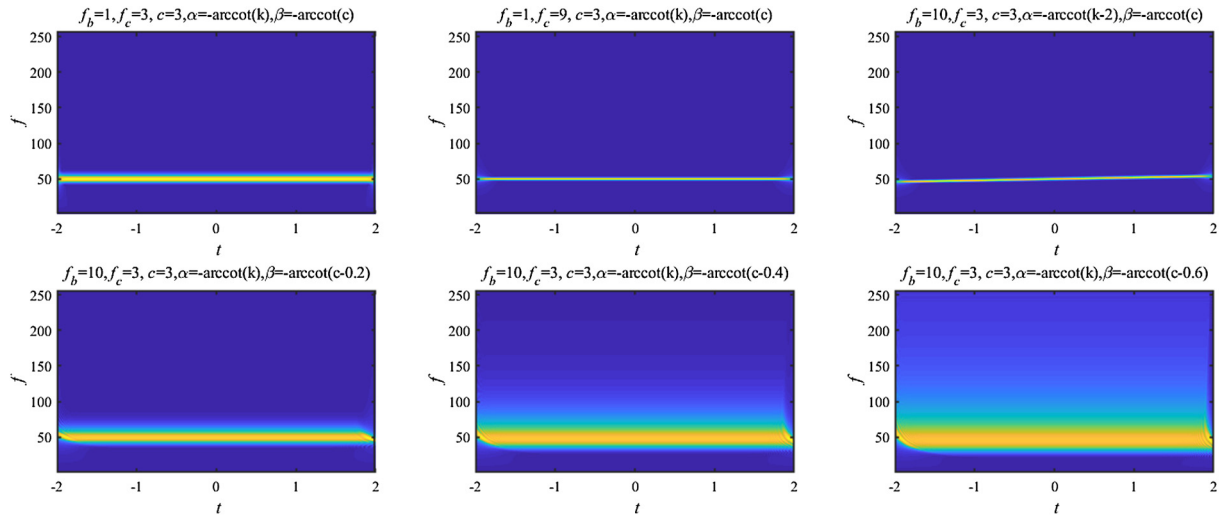


Fig. 5. The influence of FRWT parameters on LFM signal representation.

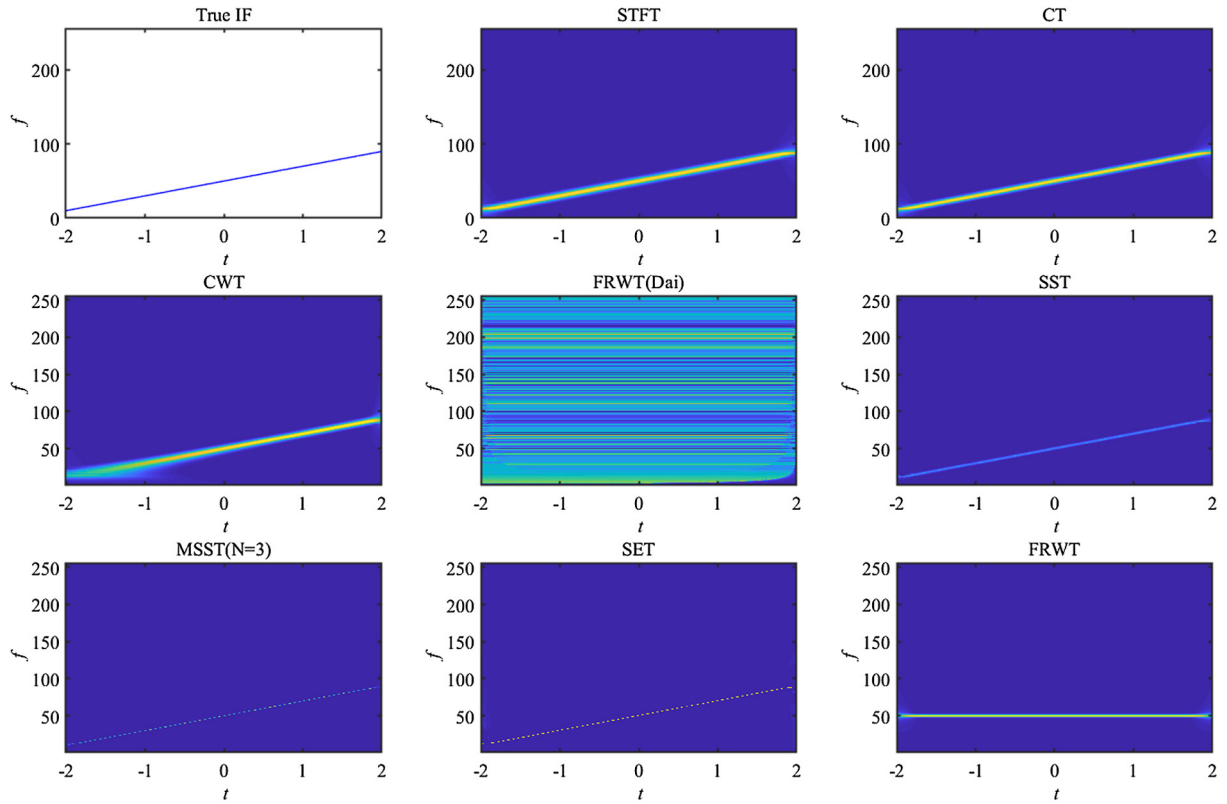


Fig. 6. Single-component LFM signal representation.

domain when $\cot \beta = -c$, and thus energy dispersion is very serious. Because α, β are both selected as optimal parameters for LFM signal and chirplet, the energy concentration of LFM signal representation by proposed FRWT is vastly superior to existing FRWT(Dai) for both pure and noisy signal; 3) Compared with three state-of-art methods (SST [20], MSST [21], SET [22]), all methods have good energy concentration for pure signal. However, the energy concentration of LFM signal representation by proposed FRWT is superior to SST, MSST, SET for noisy signal.

For a bicomponent LFM signal $f(t)$, the parameters of each component are $\{A, T, k_1, f_0, f_s\} = \{1, 4, 20, 50, 2^9\}$, $\{A, T, k_2, f_0, f_s\} = \{1, 4, 23, 50, 2^9\}$. The true IF and time-frequency representation based on five methods are shown in Fig. 8, same parameters as

the single component case are used in this simulation. The IF of two LFM signal are $IF_1(t) = 20t + 50$, $IF_2(t) = 23t + 50$, they have crossover and the slope difference is small. From Fig. 8, two LFM components can not be distinguished clearly by STFT, CT, CWT, SST, MSST and SET. However, two LFM components can be distinguished by using proposed FRWT. Another bicomponent LFM signal with parameters $\{A, T, k, f_{01}, f_s\} = \{1, 4, 20, 50, 2^9\}$ and $\{A, T, k, f_{02}, f_s\} = \{1, 4, 20, 45, 2^9\}$ is selected to test the performance of signal representation, the true IF and time-frequency representation based on five methods are shown in Fig. 9. The IF of two LFM components are parallel and close, so two LFM components are difficult to distinguish in the time-frequency plane. For this parallel bicomponent LFM signal, the two LFM components

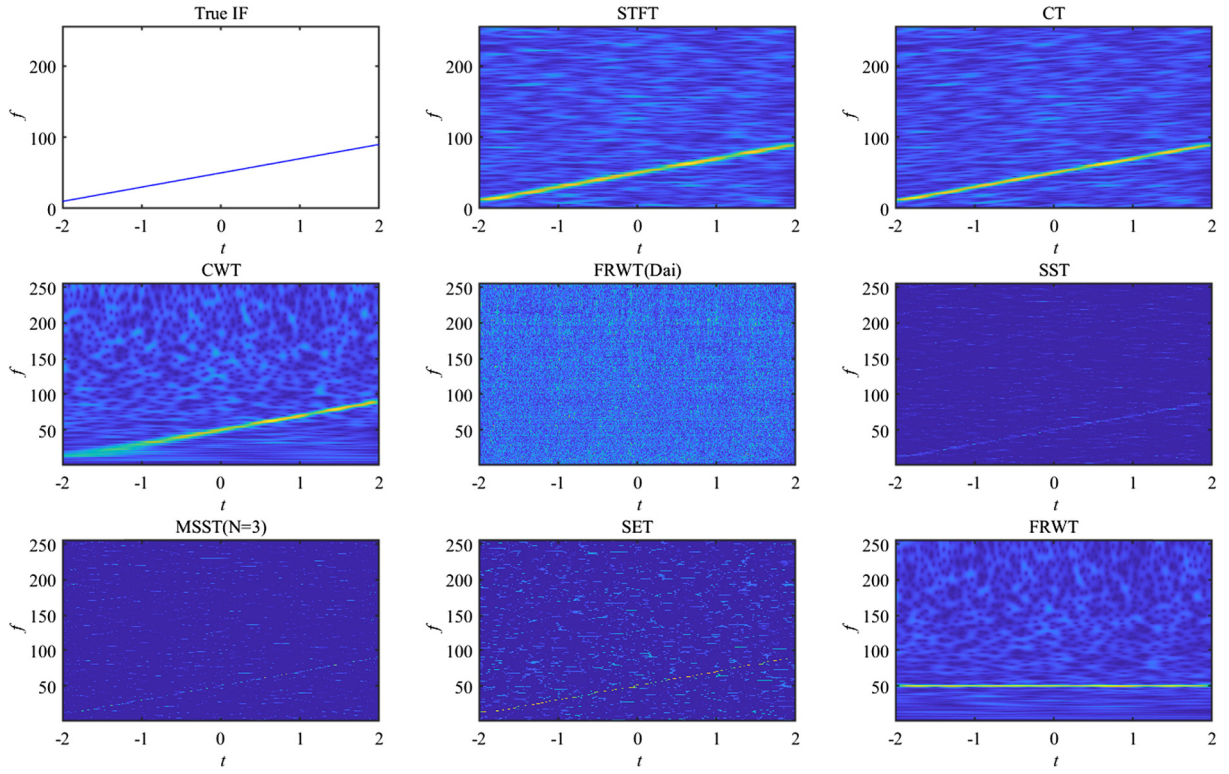


Fig. 7. Single-component noisy LFM signal representation (SNR=-5db).

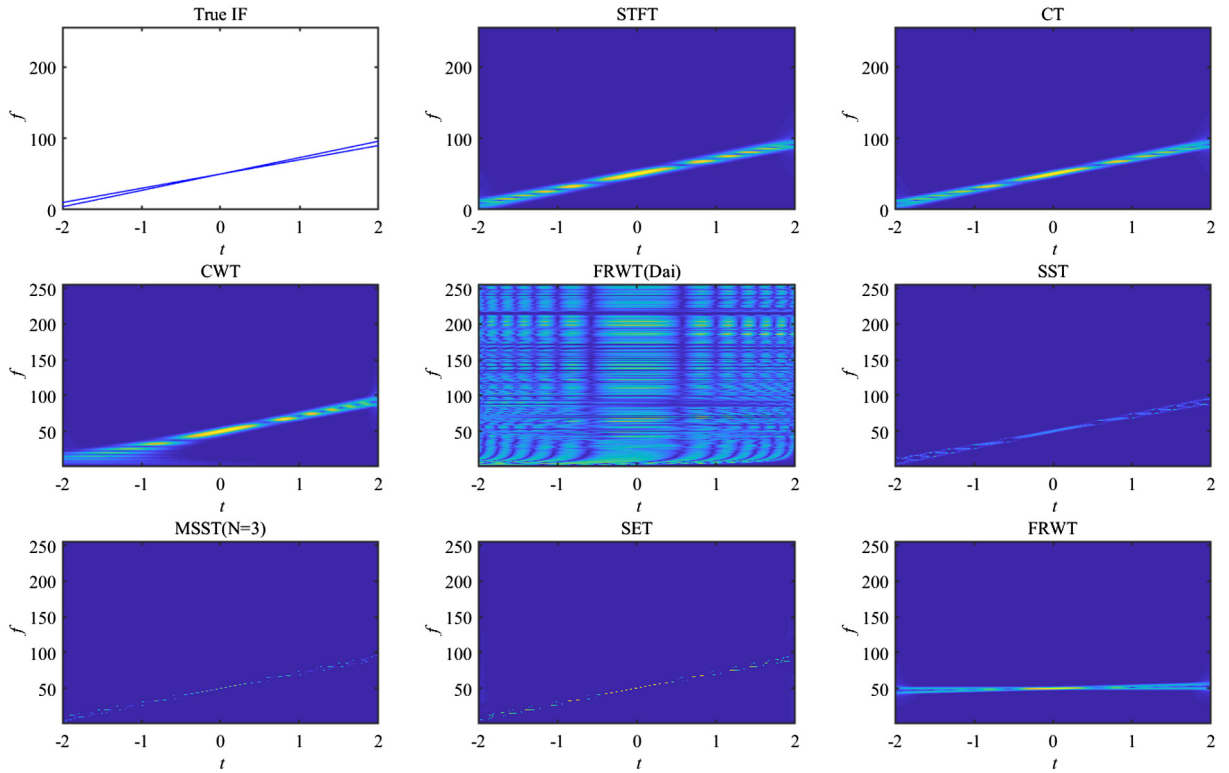


Fig. 8. Cross bicomponent LFM signal representation.

can be distinguished in the time-fractional-frequency plane. In brief, the bicomponent LFM signal time-fractional-frequency representation is better than that of some existing methods.

LFM signal representation is a common basic issue for various application research fields, all these fields involve a key point, i.e.,

energy concentration of signal representation. Owing to the better energy concentration of FRWT-based LFM signal representation method, FRWT can be further applied to some potential applications fields, such as denoising, detection and parameter estimation, separation, etc.

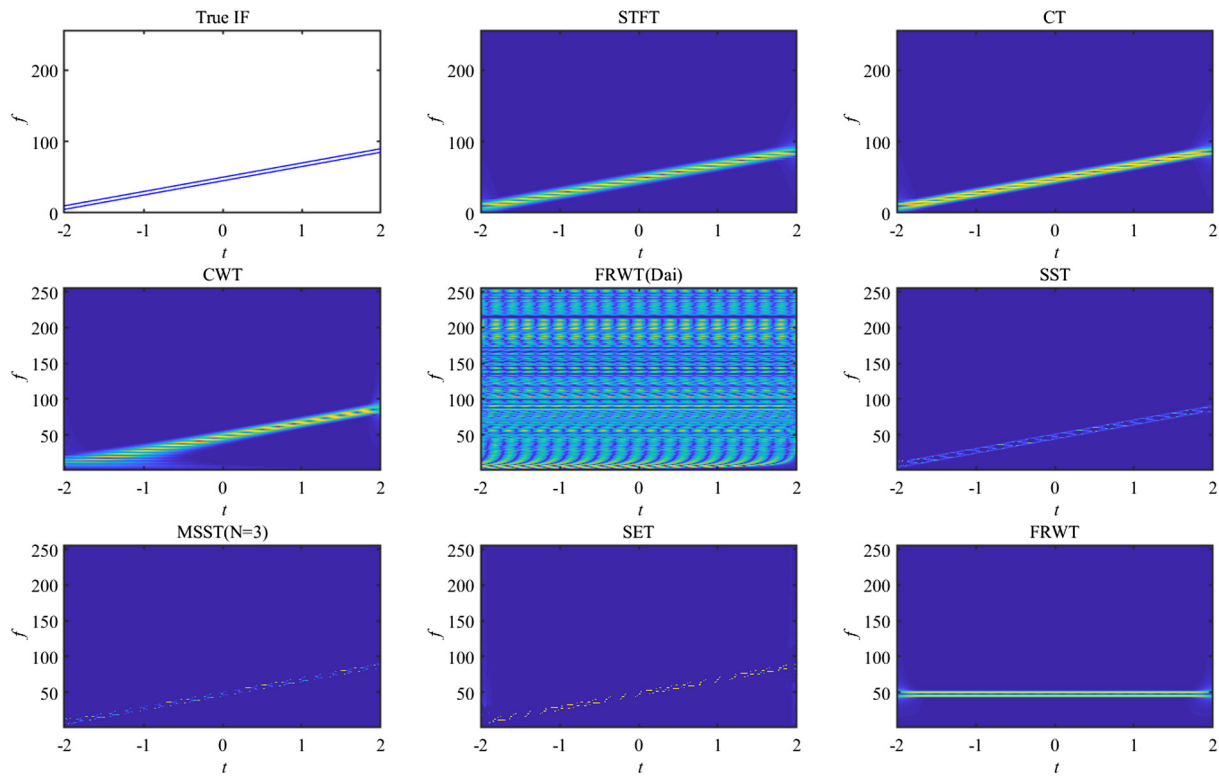


Fig. 9. Parallel bicomponent LFM signal representation.

6. Conclusion

This paper proposes a novel fractional wavelet transform to overcome the limitation of WT, this FRWT displays the time and fractional frequency information jointly in the time-fractional-frequency (TFF) plane. Some basic properties of FRWT are studied firstly, including its linear and scaling properties, inner product theorem, inverse formula and reproducing kernel. Moreover, the MRA and construction of orthogonal wavelet for FRWT are established. Finally, the application of FRWT in LFM signal TFF analysis is presented, including constant-Q property, TFF resolution, LFM signal TFF representation. The experimental results show that the energy concentration of LFM signal representation by proposed FRWT is better than that of some existing methods, which is conducive to various application researches of LFM signal.

CRedit authorship contribution statement

Yong Guo: Conceptualization, Data curation, Formal analysis, Methodology, Visualization, Writing – original draft. **Bing-Zhao Li:** Funding acquisition, Supervision, Writing – review & editing. **Li-Dong Yang:** Project administration, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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